

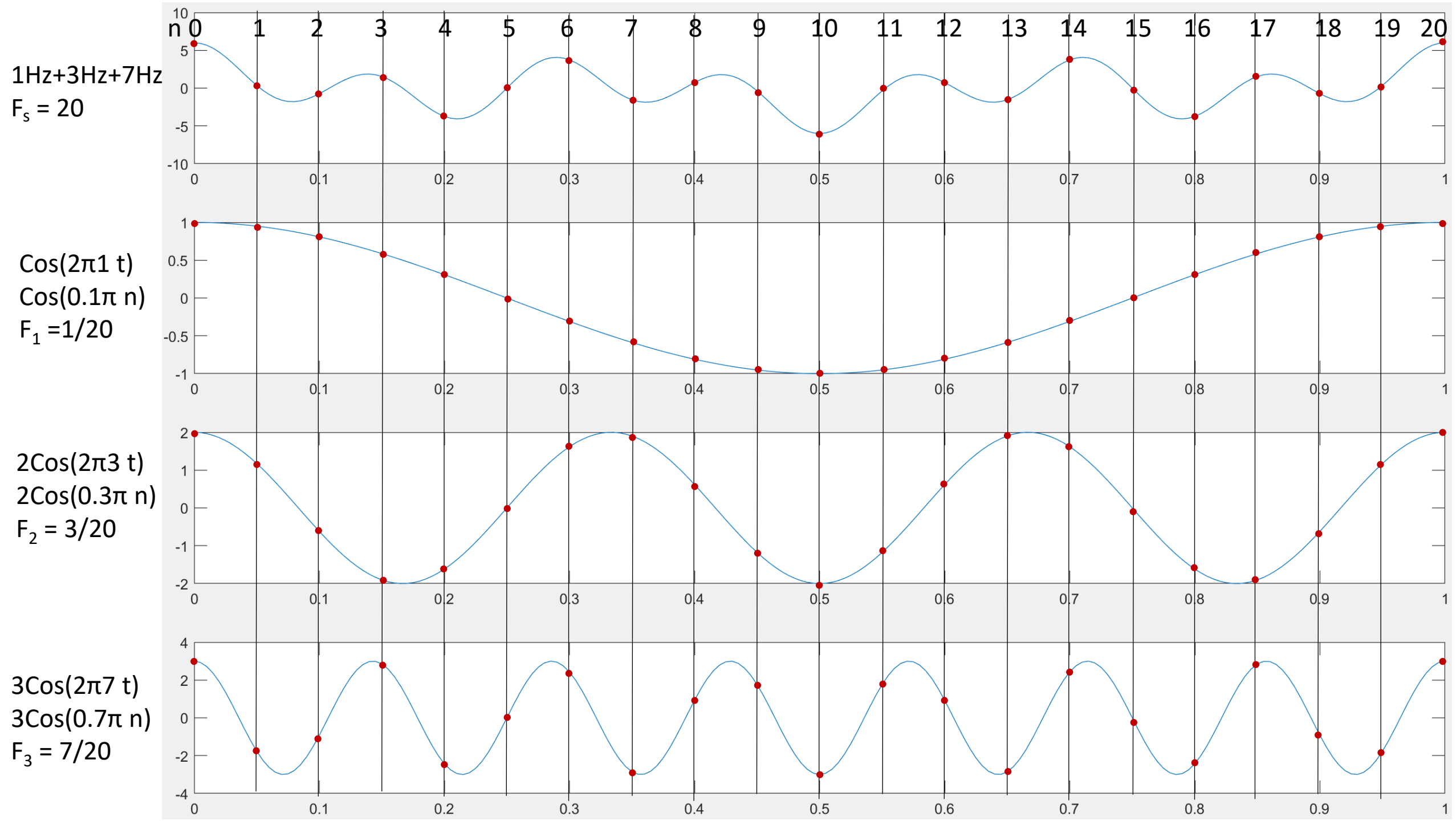
Discrete-Time Fourier Analysis

Chapter 6

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Outline

- Discrete Time Fourier Transform (DTFT)
- Properties of the DTFT
- LTID System Analysis by the DTFT
- Connection between the DTFT and CTFT
- Digital Processing of Analog Signals
- Digital Resampling: A Frequency Domain Perspective
- Generalization of the DTFT to the z-Transform



The Discrete-Time Fourier Transform (DTFT)

DTFT is a mathematical tool to represent an aperiodic discrete-time signal in terms of its frequency components (*analysis equation*), so it shows the spectrum of the signal.

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

Remember $\Omega = \frac{2\pi f}{F_s} = 2\pi F$

To find the inverse relationship, or inverse DTFT, we change the dummy variable n to m in the above equation and multiply both sides by $e^{j\Omega n}/2\pi$, and then integrate over the interval $-\pi \leq \Omega < \pi$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega)e^{j\Omega n} d\Omega$$
$$x[n] = \lim_{\Delta\Omega \rightarrow 0} \sum_{k\Delta\Omega=-\pi}^{\pi} (X(k\Delta\Omega)\Delta\mathcal{F}) e^{j(k\Delta\Omega)n}$$

This *synthesis equation* of the DTFT represents a signal $x[n]$ as a continuous sum (integral) of complex exponentials, each of which is weighted by the signal spectrum $X(\Omega)$.

Example (DTFT)

Determine the DTFT $X(\Omega)$ of a discrete-time rectangular pulse with odd length L_x ,

$$x[n] = u[n + (L_x - 1)/2] - u[n - (L_x + 1)/2].$$

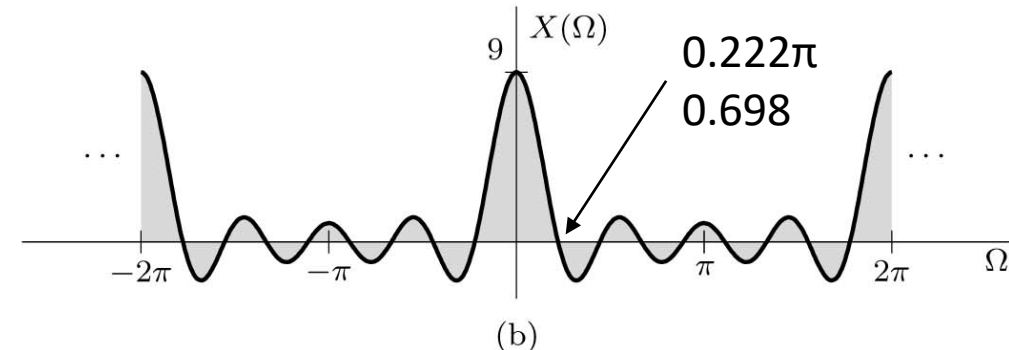
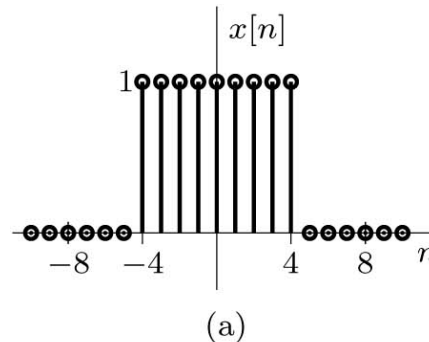
Sketch $x[n]$ and $X(\Omega)$ for $L_x = 9$.

$$\sum_{m=p}^n r^m = \frac{r^p - r^{n+1}}{1-r}$$

Solution

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

$$X(\Omega) = \frac{\sin(L_x \Omega/2)}{\sin(\Omega/2)}$$



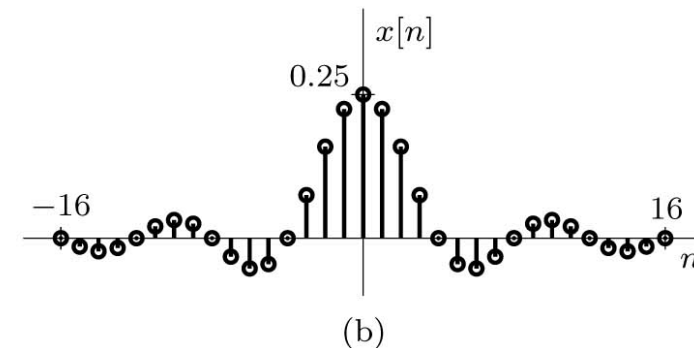
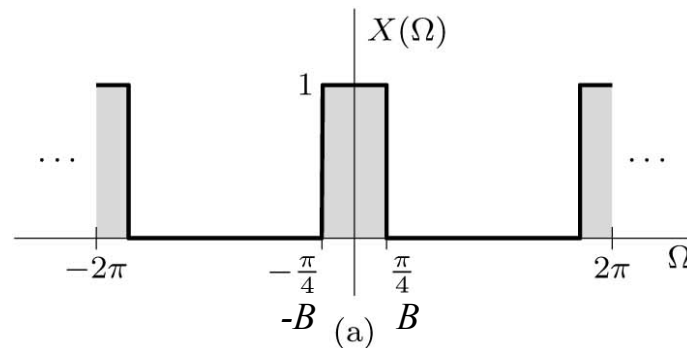
Example (IDTFT)

Determine the IDTFT $x[n]$ of the 2π -periodic rectangular pulse train described over the fundamental band $|\Omega| \leq \pi$ by $X(\Omega) = \Pi(\Omega/2B)$, where $B \leq \pi$. Sketch $X(\Omega)$ and $x[n]$ for $B = \pi/4$.

Solution

$$\text{Note: } \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

$$x[n] = \frac{B}{\pi} \text{sinc}\left(\frac{Bn}{\pi}\right)$$



Magnitude, Phase, and Existence of the DTFT

Magnitude and Phase Spectrum of $X(\Omega)$

$$X(\Omega) = |X(\Omega)|e^{j\angle X(\Omega)}$$

$$|X(\Omega)| = |X(-\Omega)| \quad \text{and} \quad \angle X(\Omega) = -\angle X(-\Omega)$$

Even
magnitude

Odd
Phase

Existence

The DTFT exist if $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$

$$x[n] = 0.8^n u[n]$$

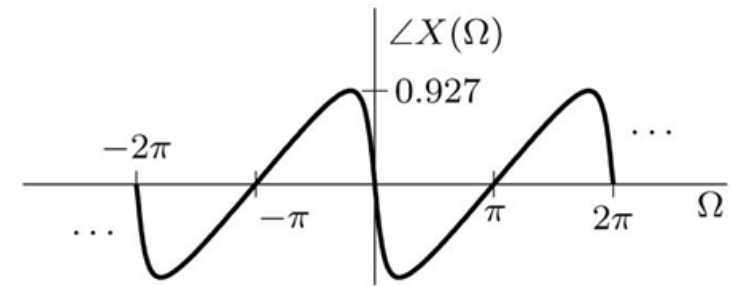
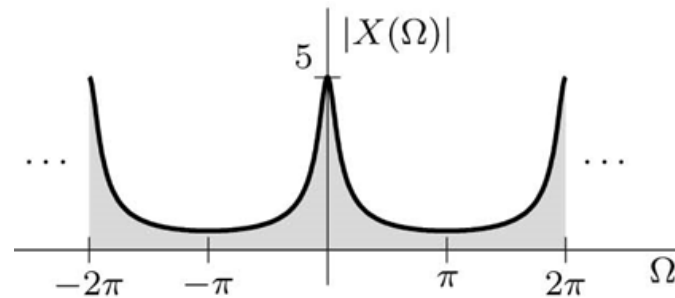
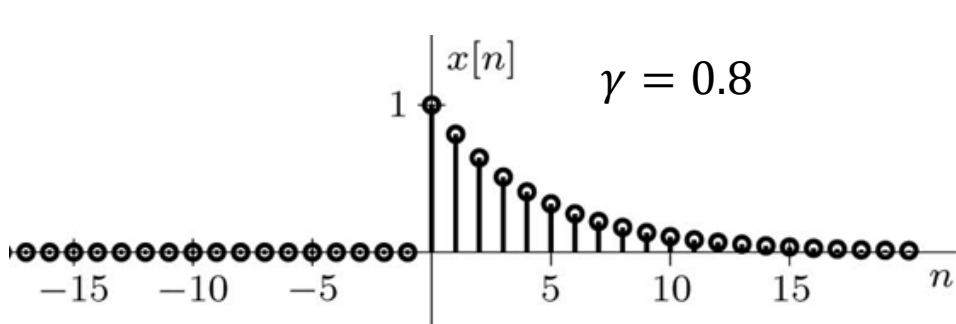
$$g[n] = 2^n u[n].$$

Example (Spectrum and Phase)

Ex1: Determine the magnitude and phase spectra of the causal exponential $x[n] = \gamma^n u[n]$.

Solution $X(\Omega) = \frac{1}{1 - \gamma e^{-j\Omega}}$ For $|\gamma| < 1$

$$|X(\Omega)| = \frac{1}{\sqrt{1 + \gamma^2 - 2\gamma \cos(\Omega)}} \quad \text{and} \quad \angle X(\Omega) = -\tan^{-1} \left(\frac{\gamma \sin(\Omega)}{1 - \gamma \cos(\Omega)} \right)$$



Obtaining the DTFT from CTFT

Synthesis CTFT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Synthesis DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega n} d\Omega$$

If the spectrum of $X(\omega)$ is bandlimited to $|\omega| < \infty$ naturally or by frequency scaling, then the DTFT can be obtained from the CTFT by replacing t by n and ω by Ω .

Example

For each of the following signals, determine the DTFT from the CTFT.

(a) $x[n] = B/\pi \operatorname{sinc}(Bn/\pi)$

(b) $x[n] = 1$

(c) $x[n] = e^{j\Omega_0 n}$

(d) $x[n] = \cos(\Omega_0 n)$

Read example in textbook

$$\frac{B}{\pi} \operatorname{sinc}\left(\frac{Bt}{\pi}\right) \stackrel{FT}{\leftrightarrow} \Pi\left(\frac{\omega}{2B}\right)$$

	$x[n]$	$X(\Omega)$	
1.	$\delta[n - k]$	$e^{-jk\Omega}$	Integer k
2.	$\gamma^n u[n]$	$\frac{e^{j\Omega}}{e^{j\Omega} - \gamma}$	$ \gamma < 1$
3.	$-\gamma^n u[-n - 1]$	$\frac{e^{j\Omega}}{e^{j\Omega} - \gamma}$	$ \gamma > 1$
4.	$\gamma^{ n }$	$\frac{1 - \gamma^2}{1 - 2\gamma \cos(\Omega) + \gamma^2}$	$ \gamma < 1$
5.	$n\gamma^n u[n]$	$\frac{\gamma e^{j\Omega}}{(e^{j\Omega} - \gamma)^2}$	$ \gamma < 1$
6.	$ \gamma ^n \cos(\Omega_0 n + \theta) u[n]$	$\frac{e^{j\Omega} [e^{j\Omega} \cos(\theta) - \gamma \cos(\Omega_0 - \theta)]}{e^{j2\Omega} - 2 \gamma \cos(\Omega_0) e^{j\Omega} + \gamma ^2}$	$ \gamma < 1$
7.	$u[n] - u[n - L_x]$	$\frac{\sin(L_x \Omega/2)}{\sin(\Omega/2)} e^{-j\Omega(L_x - 1)/2}$	
8.	$\frac{B}{\pi} \text{sinc}\left(\frac{Bn}{\pi}\right)$	$\sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{2B}\right)$	
9.	$\frac{B}{2\pi} \text{sinc}^2\left(\frac{Bn}{2\pi}\right)$	$\sum_{k=-\infty}^{\infty} \Lambda\left(\frac{\Omega - 2\pi k}{2B}\right)$	
10.	1	$2\pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$	
11.	$u[n]$	$\frac{e^{j\Omega}}{e^{j\Omega} - 1} + \pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$	
12.	$e^{j\Omega_0 n}$	$2\pi \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi k)$	
13.	$\cos(\Omega_0 n)$	$\pi \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi k) + \delta(\Omega + \Omega_0 - 2\pi k)$	
14.	$\sin(\Omega_0 n)$	$\frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi k) - \delta(\Omega + \Omega_0 - 2\pi k)$	
15.	$\cos(\Omega_0 n) u[n]$	$\frac{e^{j2\Omega} - e^{j\Omega} \cos(\Omega_0)}{e^{j2\Omega} - 2 \cos(\Omega_0) e^{j\Omega} + 1} + \frac{\pi}{2} \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi k) + \delta(\Omega + \Omega_0 - 2\pi k)$	
16.	$\sin(\Omega_0 n) u[n]$	$\frac{e^{j\Omega} \sin(\Omega_0)}{e^{j2\Omega} - 2 \cos(\Omega_0) e^{j\Omega} + 1} + \frac{\pi}{2j} \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_0 - 2\pi k) - \delta(\Omega + \Omega_0 - 2\pi k)$	

Expressing $X(\Omega)$ over the fundamental band to simplify expression by removing the summation symbol that indicated periodicity.

$x[n]$	$X(\Omega)$ for $-\pi \leq \Omega < \pi$	
1. $\delta[n - k]$	$e^{-jk\Omega}$	Integer k
2. $\gamma^n u[n]$	$\frac{e^{j\Omega}}{e^{j\Omega} - \gamma}$	$ \gamma < 1$
3. $-\gamma^n u[-n - 1]$	$\frac{e^{j\Omega}}{e^{j\Omega} - \gamma}$	$ \gamma > 1$
4. $\gamma^{ n }$	$\frac{1 - \gamma^2}{1 - 2\gamma \cos(\Omega) + \gamma^2}$	$ \gamma < 1$
5. $n\gamma^n u[n]$	$\frac{\gamma e^{j\Omega}}{(e^{j\Omega} - \gamma)^2}$	$ \gamma < 1$
6. $ \gamma ^n \cos(\Omega_0 n + \theta) u[n]$	$\frac{e^{j\Omega} [e^{j\Omega} \cos(\theta) - \gamma \cos(\Omega_0 - \theta)]}{e^{j2\Omega} - 2 \gamma \cos(\Omega_0) e^{j\Omega} + \gamma ^2}$	$ \gamma < 1$
7. $u[n] - u[n - L_x]$	$\frac{\sin(L_x \Omega/2)}{\sin(\Omega/2)} e^{-j\Omega(L_x - 1)/2}$	
8. $\frac{B}{\pi} \text{sinc}\left(\frac{Bn}{\pi}\right)$	$\Pi\left(\frac{\Omega}{2B}\right)$	$0 < B \leq \pi$
9. $\frac{B}{2\pi} \text{sinc}^2\left(\frac{Bn}{2\pi}\right)$	$\Lambda\left(\frac{\Omega}{2B}\right)$	$0 < B \leq \pi$
10. 1	$2\pi\delta(\Omega)$	
11. $u[n]$	$\frac{e^{j\Omega}}{e^{j\Omega} - 1} + \pi\delta(\Omega)$	
12. $e^{j\Omega_0 n}$	$2\pi\delta(\Omega - \Omega_0)$	$ \Omega_0 < \pi$
13. $\cos(\Omega_0 n)$	$\pi [\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$	$ \Omega_0 < \pi$
14. $\sin(\Omega_0 n)$	$\frac{\pi}{j} [\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$	$ \Omega_0 < \pi$
15. $\cos(\Omega_0 n) u[n]$	$\frac{e^{j2\Omega} - e^{j\Omega} \cos(\Omega_0)}{e^{j2\Omega} - 2\cos(\Omega_0)e^{j\Omega} + 1} + \frac{\pi}{2} [\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$	$ \Omega_0 < \pi$
16. $\sin(\Omega_0 n) u[n]$	$\frac{e^{j\Omega} \sin(\Omega_0)}{e^{j2\Omega} - 2\cos(\Omega_0)e^{j\Omega} + 1} + \frac{\pi}{2j} [\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$	$ \Omega_0 < \pi$

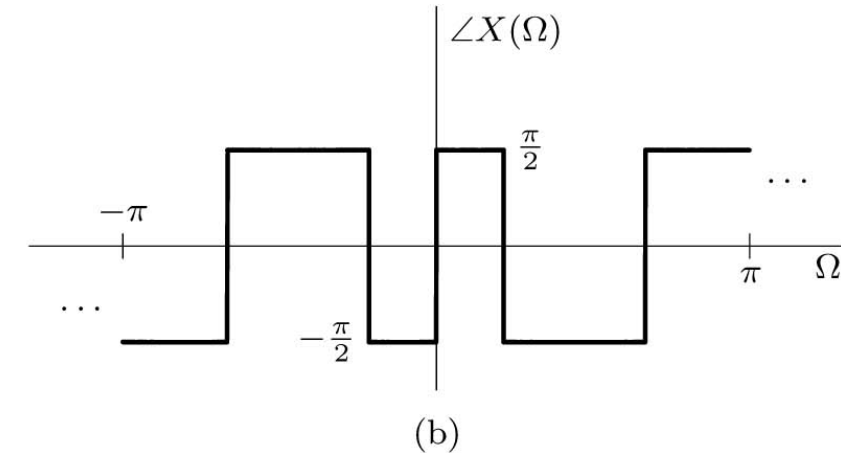
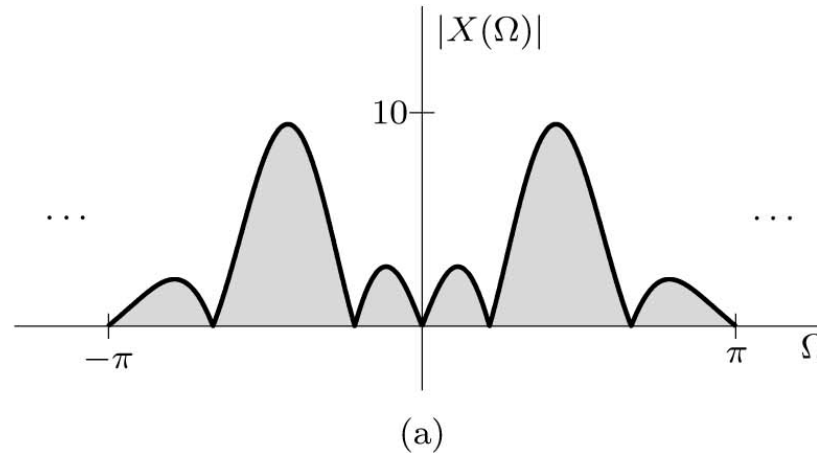
Example (DTFT of finite Duration Signals)

A signal $x[n]$ equals $[1, 2, -1, -2, 0, 2, 1, -2, -1]$ for $-4 \leq n \leq 4$ and is otherwise 0. Sketch the DTFT of this signal.

Solution

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

```
x = [1 2 -1 -2 0 2 1 -2 -1]; n2 = 4; Omega = linspace(-pi,pi,501);  
X = @(Omega) polyval(x,exp(1j*Omega))./exp(1j*Omega*n2);  
subplot(211); plot(Omega,abs(X(Omega))); subplot(212);  
plot(Omega,angle(X(Omega)));
```



6.2

Properties of the DTFT

Properties of the DTFT

Linearity Property

If $x[n] \iff X(\Omega)$ and $y[n] \iff Y(\Omega)$ then $ax[n] + by[n] \iff aX(\Omega) + bY(\Omega)$

Complex-Conjugation Property

if $x[n] \iff X(\Omega)$, then $x^*[n] \iff X^*(-\Omega)$

Time Scaling Property

No simple rule for time scaling (upsampling or downsampling).

Time-Reversal Property

if $x[n] \iff X(\Omega)$, then $x[-n] \iff X(-\Omega)$

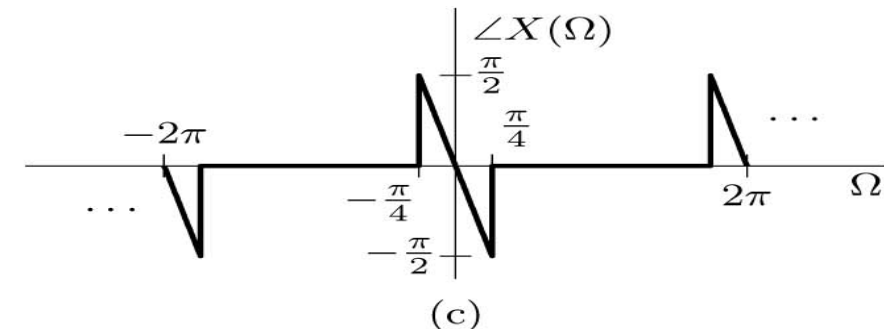
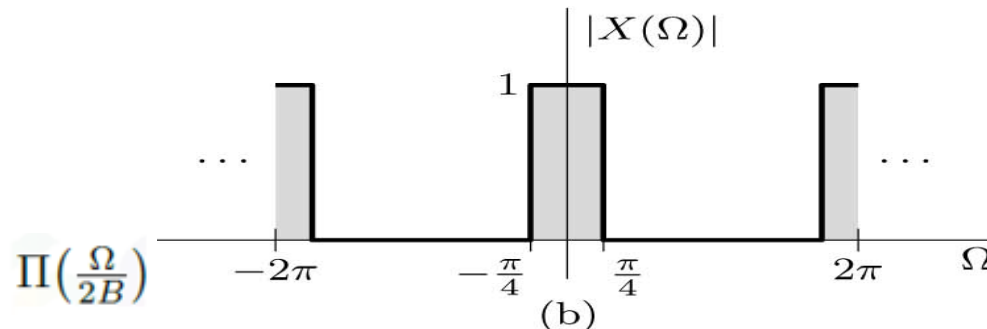
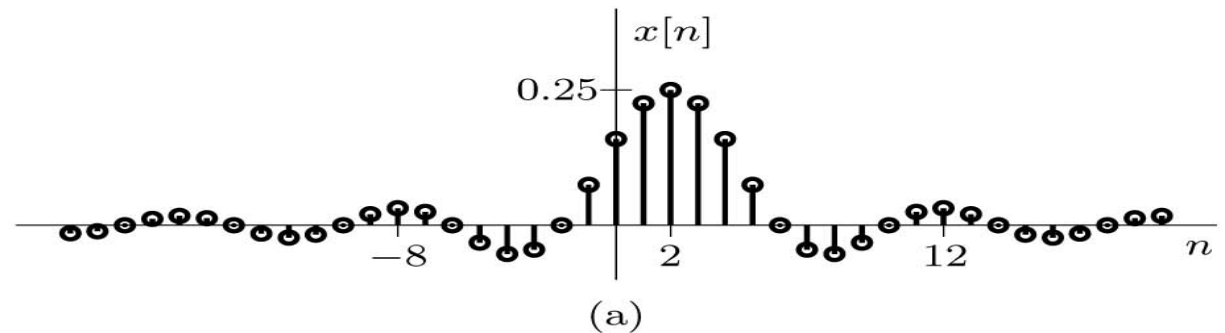
Example: Derive pair 4, $\gamma^{|n|}$, using pair 2, $\gamma^n u(n)$, and the time-reversal property

Properties of the DTFT

Time-Shifting Property

if $x[n] \iff X(\Omega)$, then $x[n - m] \iff X(\Omega)e^{-j\Omega m}$ for integer m

Example: Use the time-shifting property to determine the DTFT of $x[n] = \frac{1}{4} \text{sinc}\left(\frac{n-2}{4}\right)$



8. $\frac{B}{\pi} \text{sinc}\left(\frac{Bn}{\pi}\right)$

Properties of the DTFT

Frequency-Shifting Property

$$\text{if } x[n] \Longleftrightarrow X(\Omega), \text{ then } x[n]e^{j\Omega_0 n} \Longleftrightarrow X(\Omega - \Omega_0)$$

Modulation Property

$$x[n] \cos(\Omega_0 n) \Longleftrightarrow \frac{1}{2} [X(\Omega - \Omega_0) + X(\Omega + \Omega_0)]$$

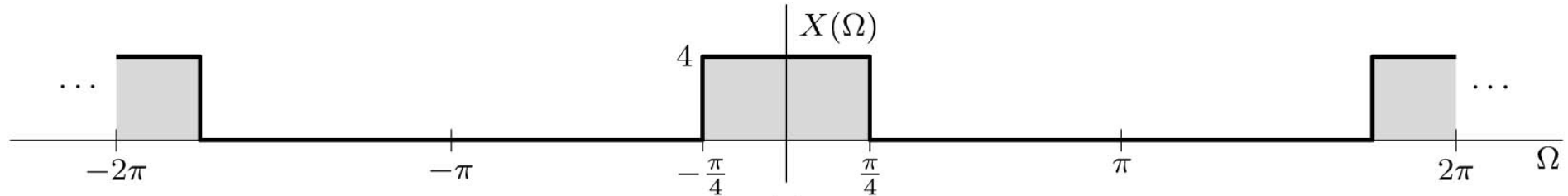
Example: A signal $x[n] = \text{sinc}(n/4)$ modulates a carrier $\cos(\Omega_0 n)$. Using the periodic expression for $X(\Omega)$ from the Table, find and sketch the spectrum of the modulated signal $x[n] \cos(\Omega_0 n)$ for (a) $\Omega_0 = 0.5\pi$ (b) $\Omega_0 = 0.85\pi$

8. $\frac{B}{\pi} \text{sinc}\left(\frac{Bn}{\pi}\right) \quad \Pi\left(\frac{\Omega}{2B}\right)$

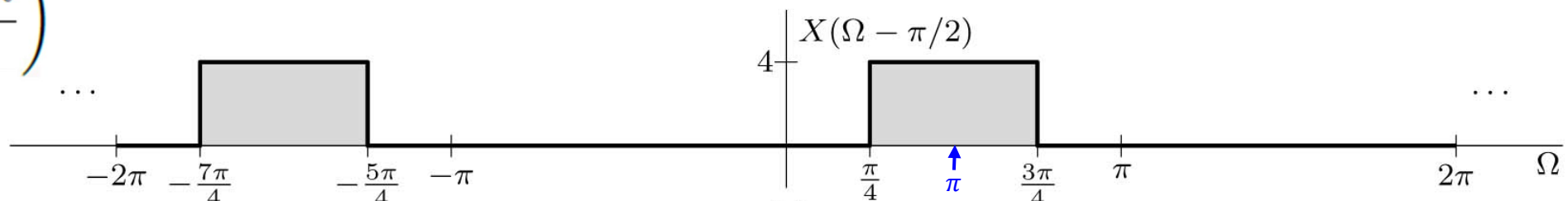
$$x[n] = \text{sinc}(n/4)$$

$$X(\Omega) = 4 \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{0.5\pi}\right)$$

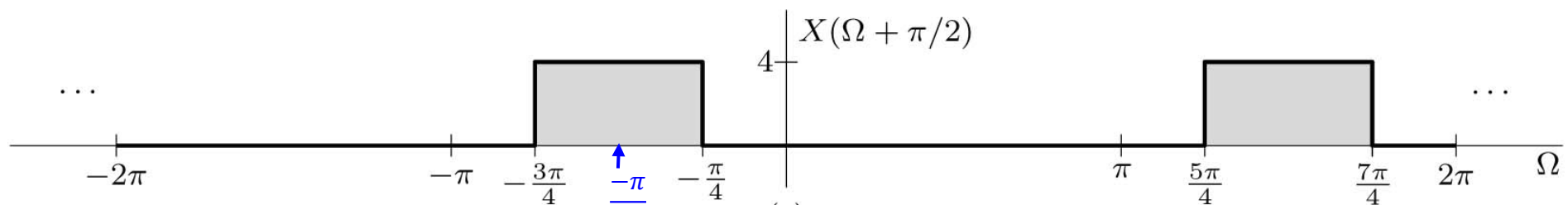
$$\Omega_o = 0.5\pi$$



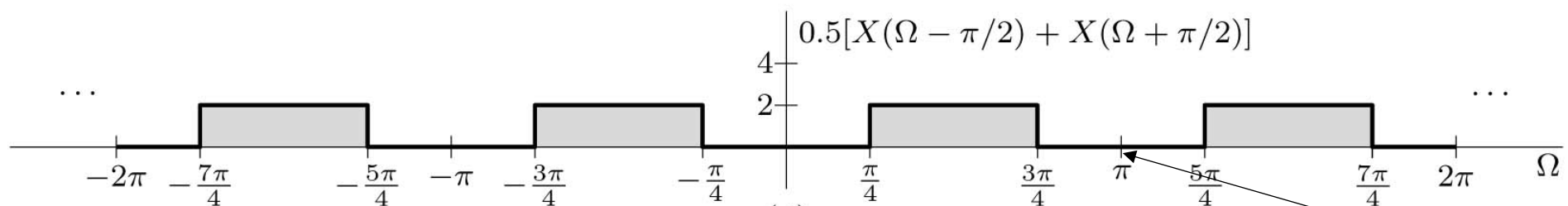
(a)



(b)



(c)

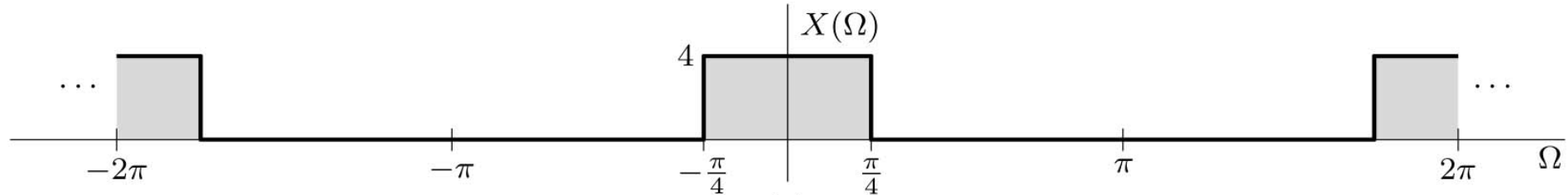


(d)

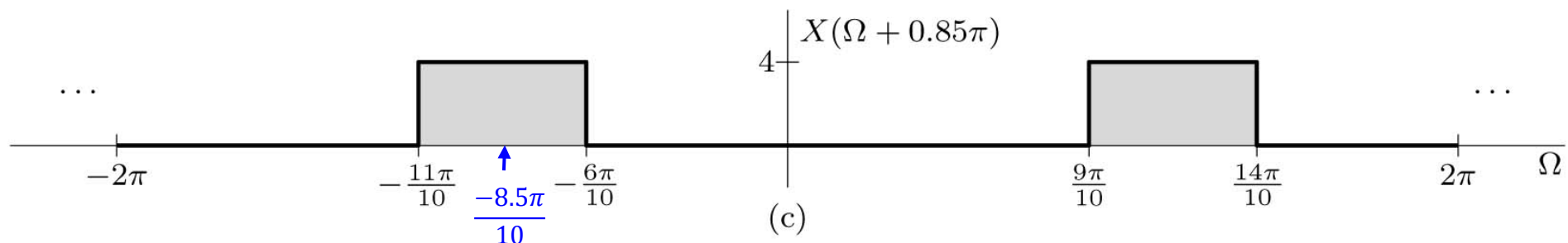
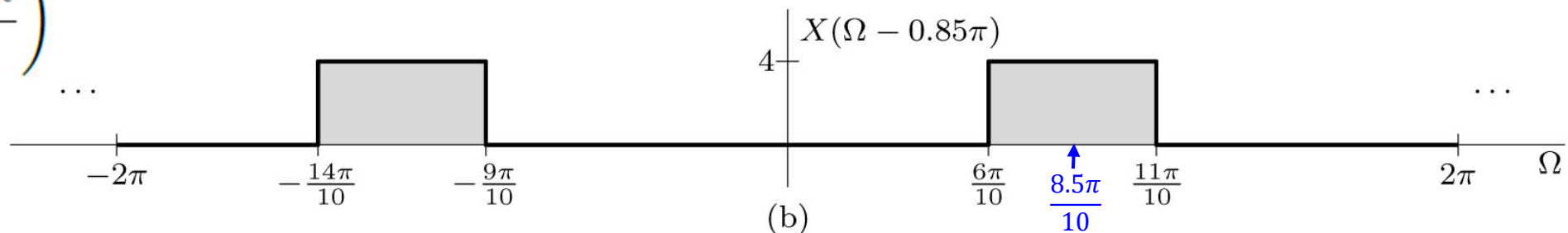
$$x[n] \cos(\pi n/2) \Longleftrightarrow 2 \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - \pi/2 - 2\pi k}{0.5\pi}\right) + \Pi\left(\frac{\Omega + \pi/2 - 2\pi k}{0.5\pi}\right)$$

Fundamental
Band

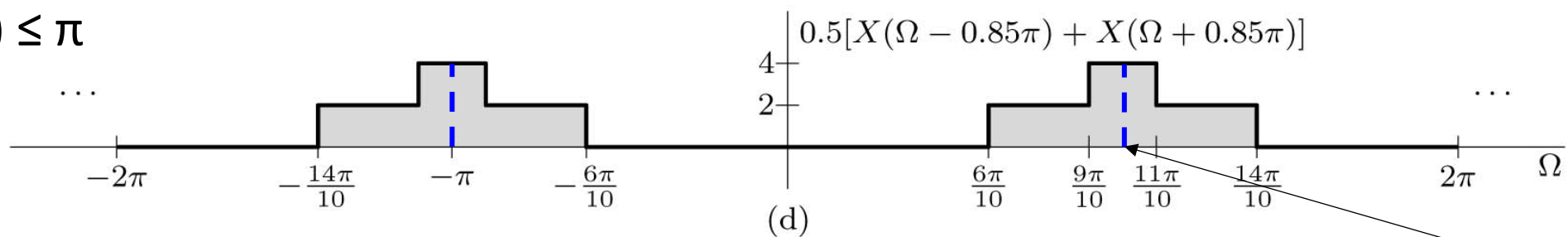
$$\Omega_0 = 0.85\pi$$



$$X(\Omega) = 4 \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{0.5\pi}\right)$$



To avoid aliasing $(\Omega_0 + B) \leq \pi$



$$x[n] \cos(0.85\pi n) \iff 2 \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 0.85\pi - 2\pi k}{0.5\pi}\right) + \Pi\left(\frac{\Omega + 0.85\pi - 2\pi k}{0.5\pi}\right)$$

Fundamental Band

Properties of the DTFT

Frequency-Differentiation Property

$$\text{if } x[n] \Longleftrightarrow X(\Omega), \text{ then } nx[n] \Longleftrightarrow j \frac{dX(\Omega)}{d\Omega}$$

Example: In the Table, derive pair 5 using pair 2 and the frequency-differentiation property.

$$\text{pair 2} \quad \gamma^n u[n] \Longleftrightarrow \frac{e^{j\Omega}}{e^{j\Omega} - \gamma}, \quad |\gamma| < 1$$

$$\text{pair 5} \quad n\gamma^n u[n] \Longleftrightarrow \frac{\gamma e^{j\Omega}}{(e^{j\Omega} - \gamma)^2}$$

Properties of the DTFT

Time-Domain Convolution Property

$$x[n] \iff X(\Omega) \quad \text{and} \quad y[n] \iff Y(\Omega)$$

$$x[n] * y[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m] \iff X(\Omega)Y(\Omega)$$

This property is the basis for the frequency-domain analysis of LTID systems.

Frequency-Domain Convolution Properties

$$x[n]y[n] \iff \frac{1}{2\pi} X(\Omega) \circledast Y(\Omega) = \frac{1}{2\pi} \int_{2\pi} X(\lambda)Y(\Omega - \lambda) d\lambda$$

Circular (or periodic) convolution

Example

Letting $x[n] = B/\pi \operatorname{sinc}(Bn/\pi)$, find and sketch the spectrum $Y(\Omega)$ of the signal $y[n] = x^2[n]$ for

(a) $0 < B \leq \pi/2$

(b) $\pi/2 < B \leq \pi$

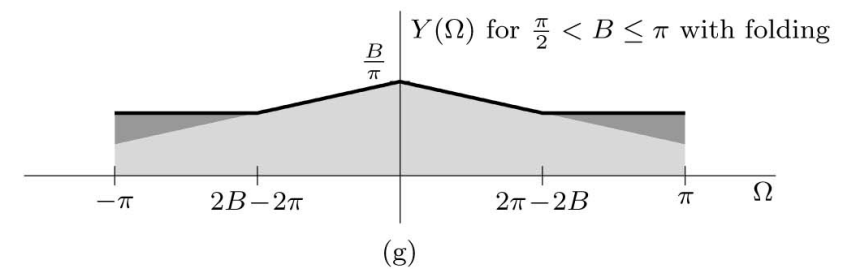
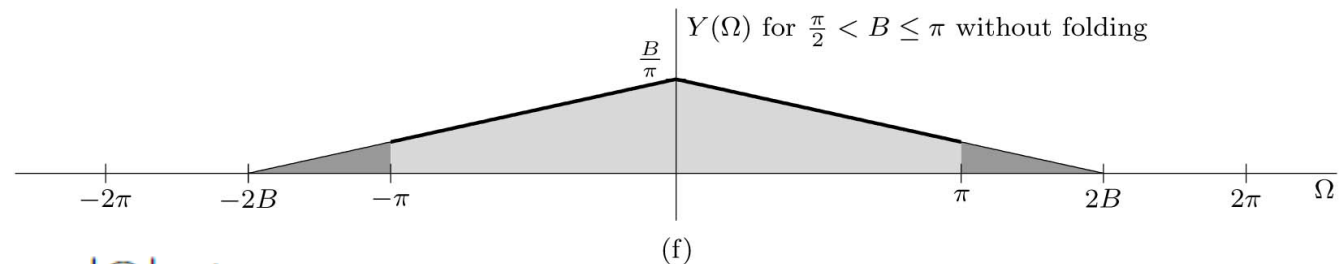
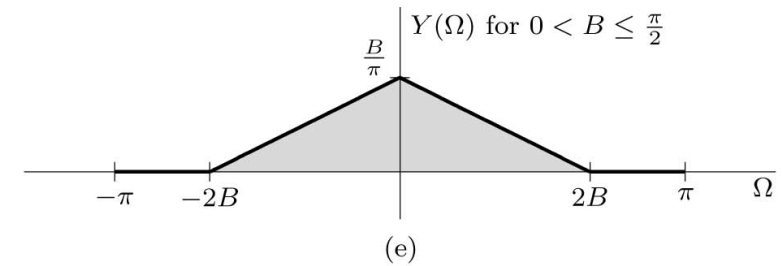
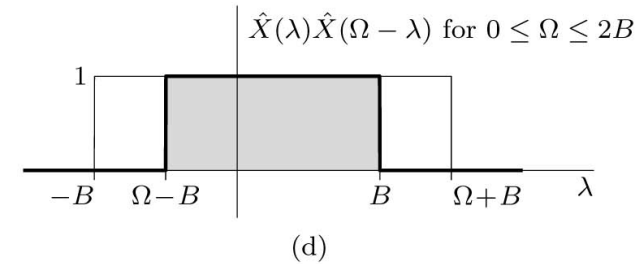
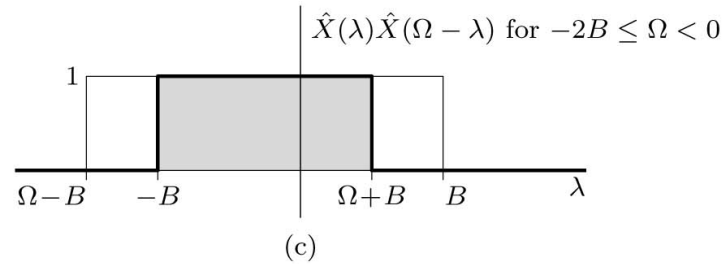
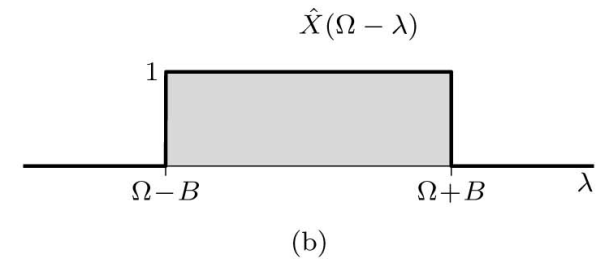
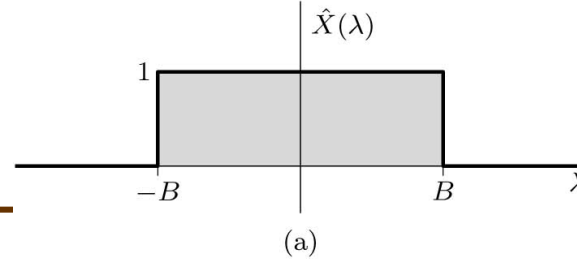
Solution

$$\frac{B}{\pi} \operatorname{sinc}\left(\frac{Bn}{\pi}\right) \Longleftrightarrow \Pi\left(\frac{\Omega}{2B}\right), \quad |\Omega| \leq \pi$$

$$\hat{X}(\Omega) * \hat{X}(\Omega) = \int_{-\pi}^{\pi} \hat{X}(\lambda) \hat{X}(\Omega - \lambda) d\lambda$$

$$x^2[n] \Longleftrightarrow \frac{1}{2\pi} \hat{X}(\Omega) * \hat{X}(\Omega) = \frac{B}{\pi} \Lambda\left(\frac{\Omega}{4B}\right), \quad |\Omega| \leq \pi$$

$$Y(\Omega) = \frac{B}{\pi} \sum_{k=-\infty}^{\infty} \Lambda\left(\frac{\Omega - 2\pi k}{4B}\right)$$



Properties of the DTFT

Correlation Property

$$8. \quad \frac{B}{\pi} \operatorname{sinc}\left(\frac{Bn}{\pi}\right) \quad \Pi\left(\frac{\Omega}{2B}\right)$$

$$\rho_{x,y}[l] = \sum_{n=-\infty}^{\infty} x[n+l]y^*[n]$$

$$\rho_{x,y}[l] = x[l] * y^*[-l] \iff X(\Omega)Y^*(\Omega)$$

Finding Signal Energy in the Frequency-Domain

$$\rho_{x,x}[0] = \sum_{n=-\infty}^{\infty} x[n]x^*[n] = \sum_{n=-\infty}^{\infty} |x[n]|^2 = E_x$$

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(\Omega)|^2 d\Omega = \frac{1}{\pi} \int_0^\pi |X(\Omega)|^2 d\Omega \quad \text{Parseval's Theorem}$$

Example: Use Parseval's theorem to find the energy of $x[n] = \operatorname{sinc}(Bn/\pi)$.

Discrete-Time Fourier Transform

Synthesis:

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(\Omega) e^{j\Omega n} d\Omega$$

Analysis:

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

Duality:

Linearity:

$$ax[n] + by[n] \iff aX(\Omega) + bY(\Omega)$$

Complex Conjugation:

$$x^*[n] \iff X^*(-\Omega)$$

Scaling and Reversal:

(see Sec. 6.6)

$$x[-n] \iff X(-\Omega)$$

Shifting:

$$x[n-m] \iff X(\Omega) e^{-j\Omega m}$$

$$x[n] e^{j\Omega_0 n} \iff X(\Omega - \Omega_0)$$

Differentiation:

$$-jnx[n] \iff \frac{d}{d\Omega} X(\Omega)$$

Time Integration:

Convolution:

$$x[n] * y[n] \iff X(\Omega) Y(\Omega)$$

$$x[n] y[n] \iff \frac{1}{2\pi} X(\Omega) \circledast Y(\Omega)$$

Correlation:

$$\rho_{x,y}[l] = x[l] * y^*[-l] \iff X(\Omega) Y^*(\Omega)$$

Parseval's:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(\Omega)|^2 d\Omega$$

Fourier Transform

Synthesis:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Analysis:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Duality:

$$\text{if } x(t) \iff X(\omega), \text{ then } X(t) \iff 2\pi x(-\omega)$$

Linearity:

$$ax(t) + by(t) \iff aX(\omega) + bY(\omega)$$

Complex Conjugation:

$$x^*(t) \iff X^*(-\omega)$$

Scaling and Reversal:

$$x(at) \iff \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$x(-t) \iff X(-\omega)$$

Shifting:

$$x(t-t_0) \iff X(\omega) e^{-j\omega t_0}$$

$$x(t) e^{j\omega_0 t} \iff X(\omega - \omega_0)$$

Differentiation:

$$\frac{d}{dt} x(t) \iff j\omega X(\omega)$$

$$-jtx(t) \iff \frac{d}{d\omega} X(\omega)$$

Time Integration:

$$\int_{-\infty}^t x(\tau) d\tau \iff \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$$

Convolution:

$$x(t) * y(t) \iff X(\omega) Y(\omega)$$

$$x(t) y(t) \iff \frac{1}{2\pi} X(\omega) \circledast Y(\omega)$$

Correlation:

$$\rho_{x,y}(\tau) = x(\tau) * y^*(-\tau) \iff X(\omega) Y^*(\omega)$$

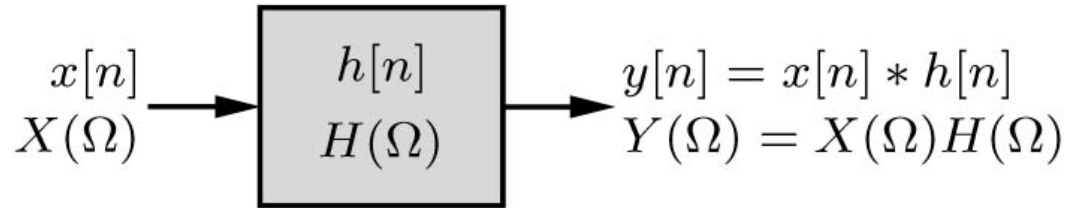
Parseval's:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

6.3

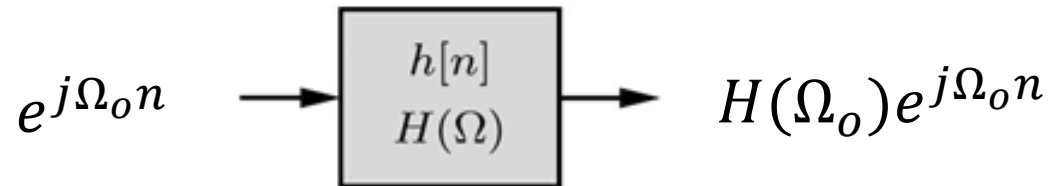
LTID System Analysis by the DTFT

LTID System Analysis



$$y[n] = \frac{1}{2\pi} \int_{2\pi} X(\Omega)H(\Omega)e^{j\Omega n} d\Omega$$

The output $y[n]$ is a scaled sum of the responses to all the component frequencies of the input.



If the input is everlasting exponential, $x[n] = e^{j\Omega_o n}$, then the output will be the same as the input multiplied by the system response to the frequency Ω_o , $y[n] = H(\Omega_o)e^{j\Omega_o n}$.

Frequency Response from a Difference Equation

$$\sum_{k=0}^K a_k y[n-k] = \sum_{l=0}^K b_l x[n-l] \quad \Longleftrightarrow \quad \sum_{k=0}^K a_k Y(\Omega) e^{-jk\Omega} = \sum_{l=0}^K b_l X(\Omega) e^{-jl\Omega}$$

$$\left(\sum_{k=0}^K a_k e^{-jk\Omega} \right) Y(\Omega) = \left(\sum_{l=0}^K b_l e^{-jl\Omega} \right) X(\Omega)$$

$$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{\sum_{l=0}^K b_l e^{-jl\Omega}}{\sum_{k=0}^K a_k e^{-jk\Omega}}$$

Find $H(\Omega)$ and determine the zero-state response $y[n]$ to input $x[n] = \cos(3\pi n/10)$ of an LTID system described by the difference equation

$$y[n] - 1.8y[n-1] + 0.81y[n-2] = x[n] + 2x[n-1]$$

Solution

$$y[n] = 3.61 \cos(3\pi n/10 - 2.63)$$

Example

An LTID system is specified by the equation

$$y[n] - 0.5y[n-1] = x[n].$$

Find $H(\Omega)$, the frequency response of this system, and determine the zero-state response $y[n]$ if the input $x[n] = (0.8)^n u[n]$.

Solution

$$y[n] = \left[\frac{8}{3}(0.8)^n - \frac{5}{3}(0.5)^n \right] u[n]$$

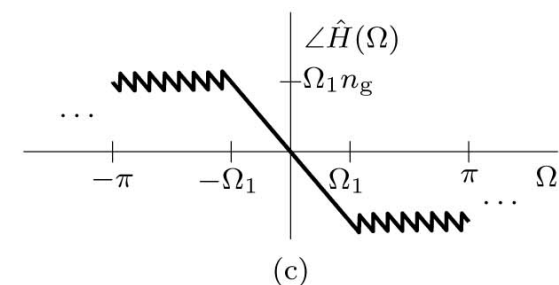
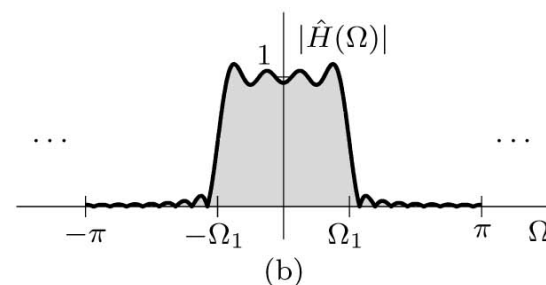
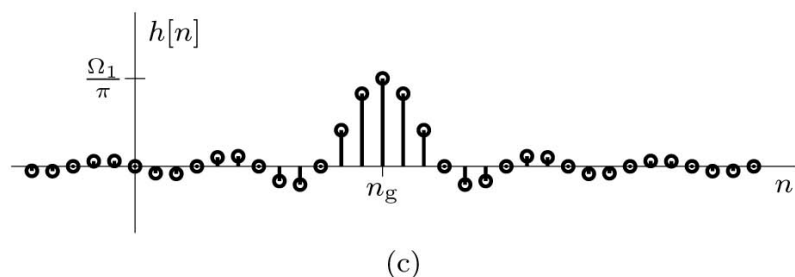
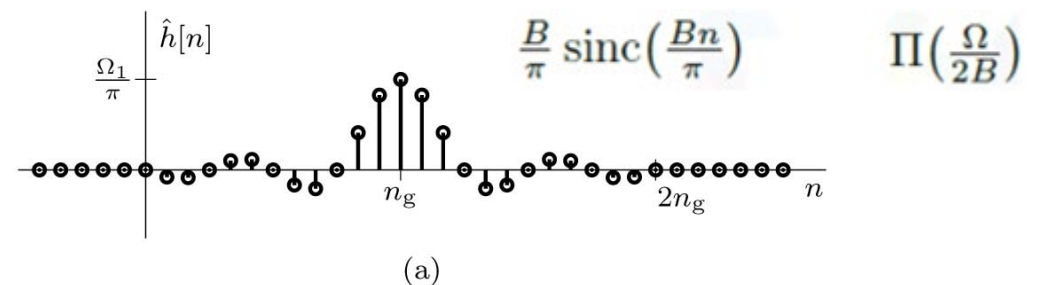
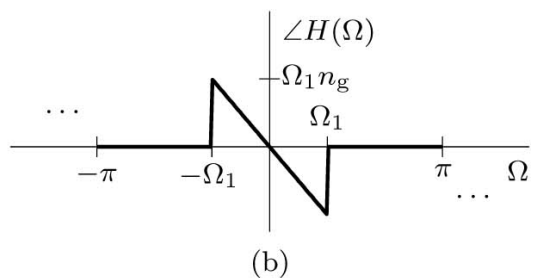
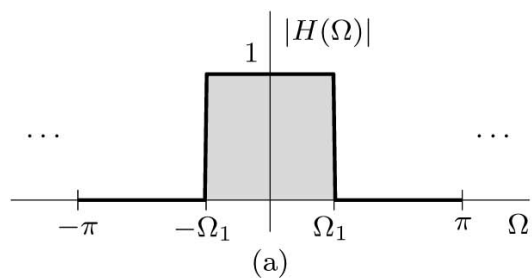
$$\gamma^n u[n]$$

$$\frac{e^{j\Omega}}{e^{j\Omega} - \gamma}$$

Matlab Code to find Partial Fraction Expansion

```
[r,p,k] = residuez(1,poly([0.8,0.5]))  
r = 2.6667 -1.6667  
p = 0.8000 0.5000  
k = []
```

Ideal and Realizable Filters



$$H(\Omega) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{2\Omega_1}\right) e^{-j\Omega n_g}$$

$$\hat{h}[n] = h[n] \Pi\left(\frac{n - n_g}{2n_g^+}\right)$$

$$h[n] = \frac{\Omega_1}{\pi} \text{sinc}\left[\frac{\Omega_1(n - n_g)}{\pi}\right]$$

$$\hat{H}(\Omega) = \sum_{n=-\infty}^{\infty} \hat{h}[n] e^{-j\Omega n}$$

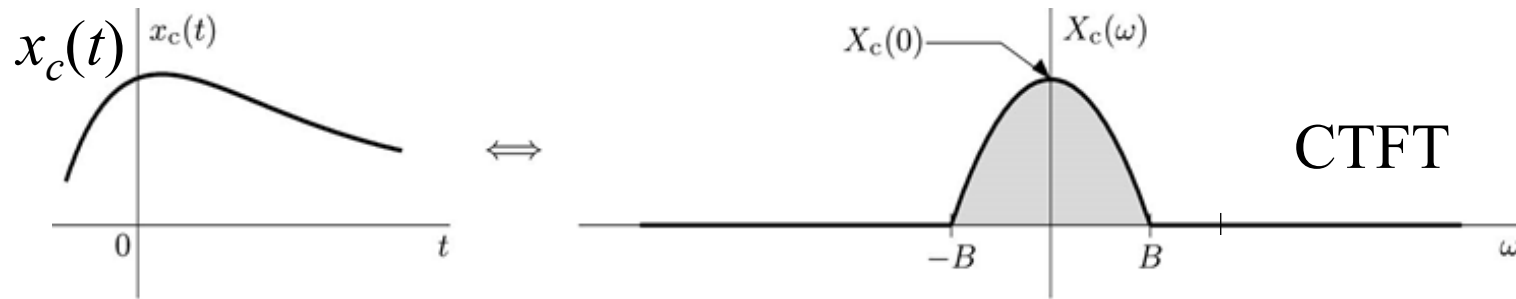
Ideal Lowpass Filter: Noncausal System

Realizable Lowpass Filter: with delay and ripple in pass band region.

6.4

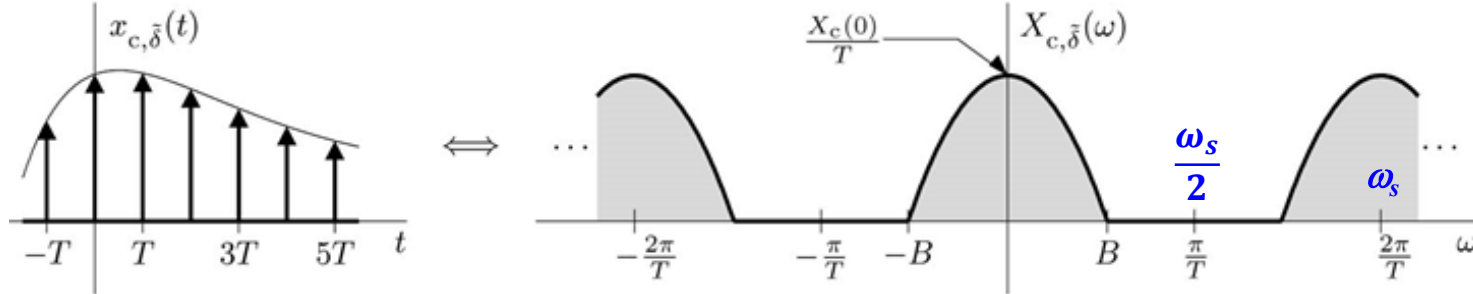
Connection between the DTFT and the CTFT

Connection between the DTFT and the CTFT



$$T = 1/f_s \quad \omega_s = 2\pi/T$$

Impulse sampling

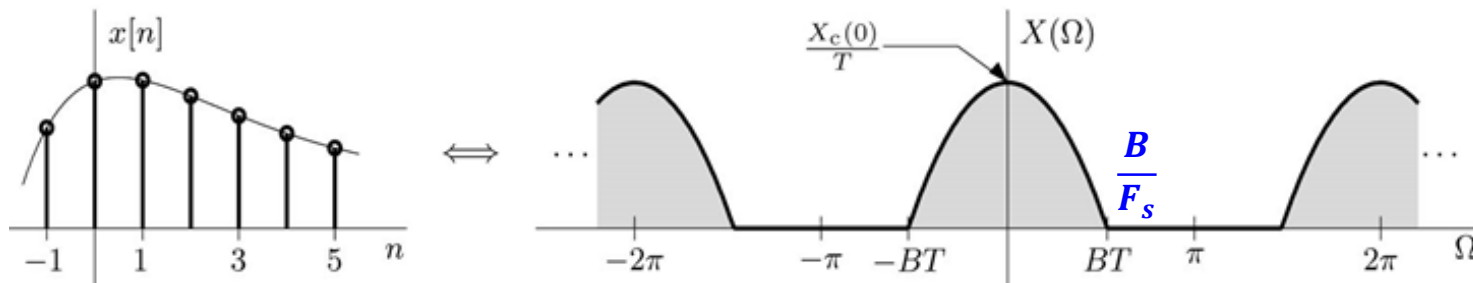


$$X(\omega T) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\omega - k\omega_s)$$

$$\Omega = \omega T$$

Replace ω
with Ω/T

Point sampling



DTFT

$$X(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(\frac{\Omega - 2\pi k}{T}\right)$$

$$X(\Omega) = \frac{1}{T} X_c\left(\frac{\Omega}{T}\right)$$

The DTFT of a DT signal $x[n]$ equals a scaled periodic replication of the CTFT of the CT signal $x(t)$ sampled at T .

Example

Consider a bandlimited signal $x_c(t)$ with spectrum

$$X_c(\omega) = \begin{cases} \frac{100}{\omega^2 + 10^4} & |\omega| < 50\pi \\ 0 & \text{otherwise} \end{cases}$$

For $T = 0.01$, determine the DTFT $X(\Omega)$ of $x[n] = x_c(nT)$. Discuss what happens if instead $T = 0.1$.

$$X(\Omega) = \frac{1}{T} X_c\left(\frac{\Omega}{T}\right)$$

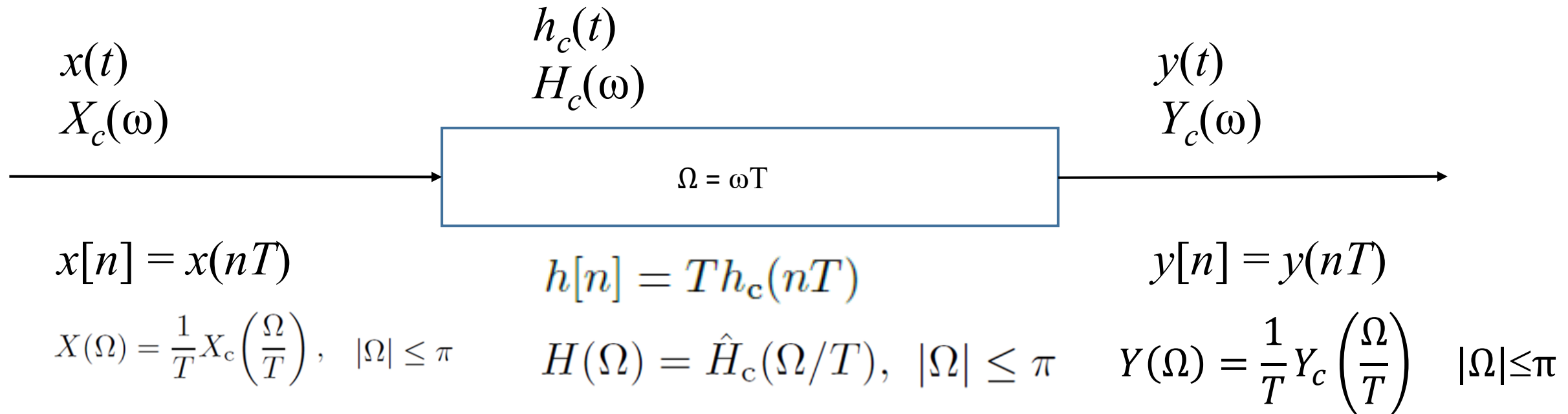
Answer

$$X(\Omega) = \begin{cases} \frac{1}{\Omega^2 + 1} & |\Omega| < 0.5\pi \\ 0 & \text{otherwise} \end{cases}$$

$$X(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(\frac{\Omega - 2\pi k}{T}\right)$$

6.5

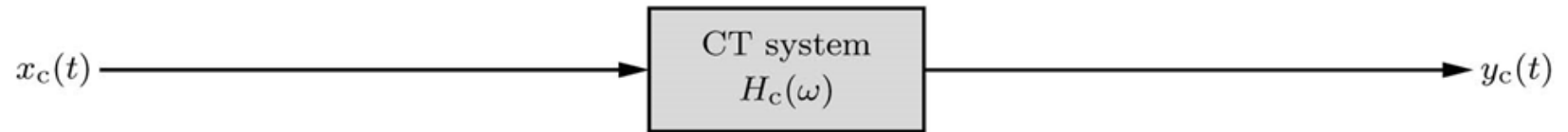
Digital Processing of Analog Signals



Digital Processing of Analog Signals

Important conditions for the impulse invariance method: (design of discrete IIR from continuous)

1. The sampling rate is above the Nyquist rate, higher than twice the bandwidth of the signal $x(t)$
2. The system is bandlimited



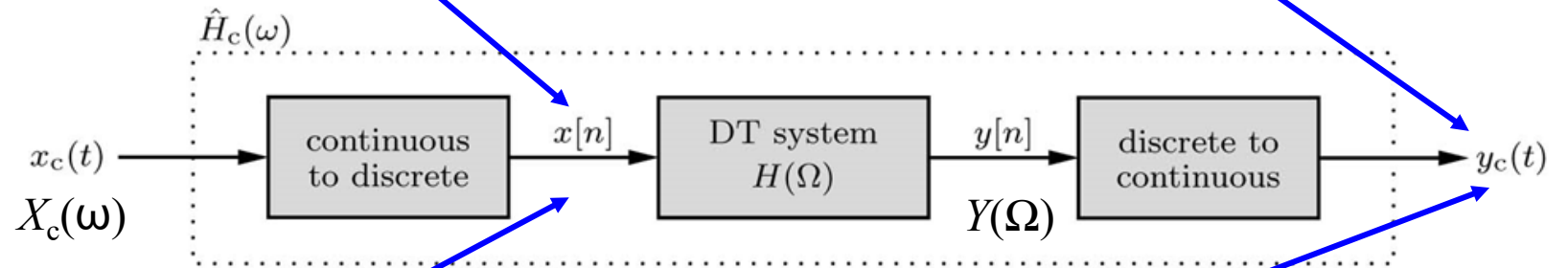
Time domain Characterization of ideal C/D and D/C Converter

$$\hat{H}_c(\omega) = \begin{cases} H_c(\omega) & |\omega| \leq \omega_s/2 \\ 0 & \text{otherwise} \end{cases}$$

$$x[n] = x_c(nT)$$

$$y_c(t) = \sum_{n=-\infty}^{\infty} y[n] \operatorname{sinc}\left(\frac{t - nT}{T}\right)$$

$$H(\Omega) = \hat{H}_c(\Omega/T), \quad |\Omega| \leq \pi$$



$$X(\Omega) = \frac{1}{T} X_c\left(\frac{\Omega}{T}\right), \quad |\Omega| \leq \pi \quad Y_c(\omega) = T Y(\omega T), \quad |\omega T| \leq \pi$$

Frequency domain Characterization of ideal C/D and D/C Converter

Digital Filter Design from Continuous Filter $H_c(\omega)$

Frequency-domain criterion for digital filter design:

$$H(\Omega) = H_c(\Omega/T), \quad |\Omega| \leq \pi.$$

Prove

$$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{\frac{1}{T} Y_c(\Omega/T)}{\frac{1}{T} X_c(\Omega/T)} = H_c(\Omega/T)$$

Time-domain criterion for digital filter design (IIR impulse invariance method):

$$h[n] = T h_c(nT)$$

Prove

$$h_c(t) = \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} H_c(\omega) e^{j\omega t} d\omega \xrightarrow[\omega \rightarrow \Omega/T]{t \rightarrow nT} h_c(nT) = \frac{1}{2\pi T} \int_{-\pi}^{\pi} H(\Omega) e^{j\Omega n} d\Omega = \frac{1}{T} h[n]$$

Example

Using the minimum possible sampling frequency, design a DT system that delays a bandlimited ($B \leq 10$ kHz) CT input $x_c(t)$ by $\tau = 250 \mu\text{s}$. Sketch the frequency responses $H(\Omega)$ and $\hat{H}_c(\omega)$. Explain how closely the system compares with an ideal analog delay $\tau = 250 \mu\text{s}$. Using a hypothetical bandlimited ($B \leq 10$ kHz) signal $x_c(t)$ and its spectrum $X_c(\omega)$, sketch the signals and spectra at the input of the C/D converter, the input of $H(\Omega)$, the output of $H(\Omega)$, and the output of the D/C converter.

Answer

$$y_c(t) = x_c(t - \tau)$$

$$H_c(\omega) = e^{-j\omega\tau}$$

$$H(\Omega) = H_c(\Omega/T)$$

$$H(\Omega) = e^{-j\Omega\tau/T} = e^{-j5\Omega} \quad |\Omega| \leq \pi$$

$$H_c(\omega) = e^{-j\omega\tau} = e^{-j5\omega T}$$

$$H(\Omega) = H_c(\Omega/T) = e^{-j5\Omega}, \quad |\Omega| \leq \pi$$

$$h[n] = \delta[n - 5]$$

$$\hat{H}_c(\omega) = \begin{cases} e^{-j5\omega T} & |\omega T| \leq \pi \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{H}_c(\omega) = e^{-j\omega/4000}$$

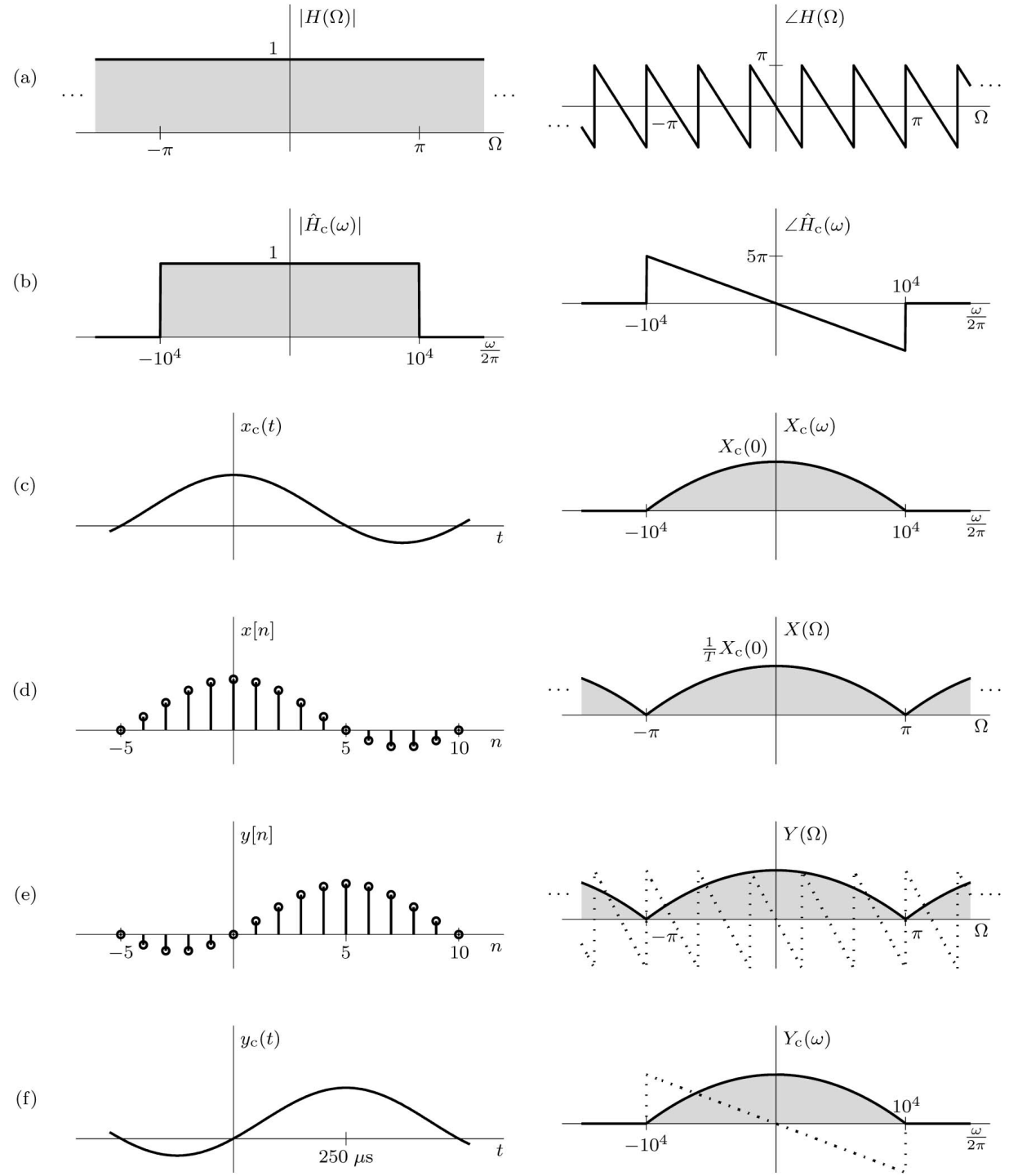
$$X(\Omega) = \frac{1}{T} X_c(\Omega/T), \quad |\Omega| \leq \pi$$

$$Y(\Omega) = X(\Omega)H(\Omega) = X(\Omega) e^{-j5\Omega}$$

$$y[n] = x[n - 5]$$

$$Y_c(\omega) = X_c(\omega) e^{-j5\omega T}$$

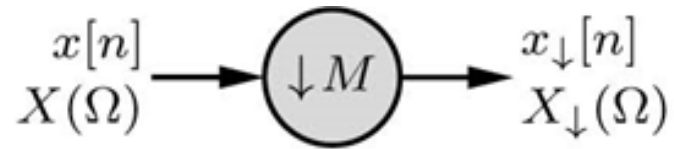
$$y_c(t) = x_c(t - 5T) = x_c(t - \tau), \quad \tau = 250 \mu s.$$



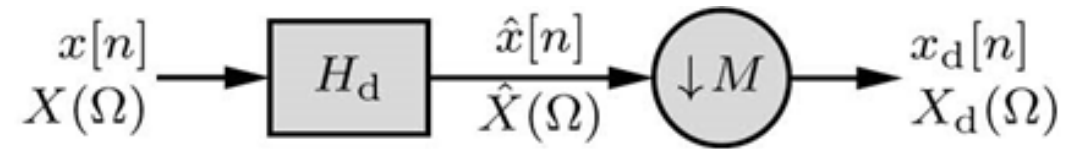
6.6

Digital Resampling:
A Frequency-Domain Perspective

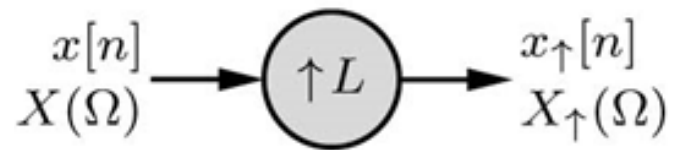
Digital Resampling: A Frequency-Domain Perspective



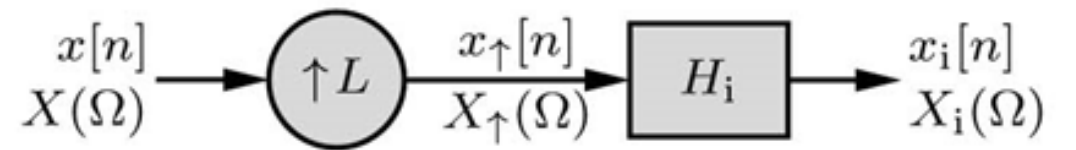
Downsampling



Decimation

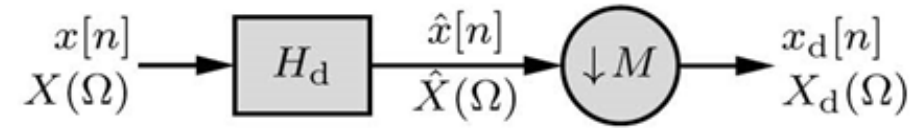


Upsampling

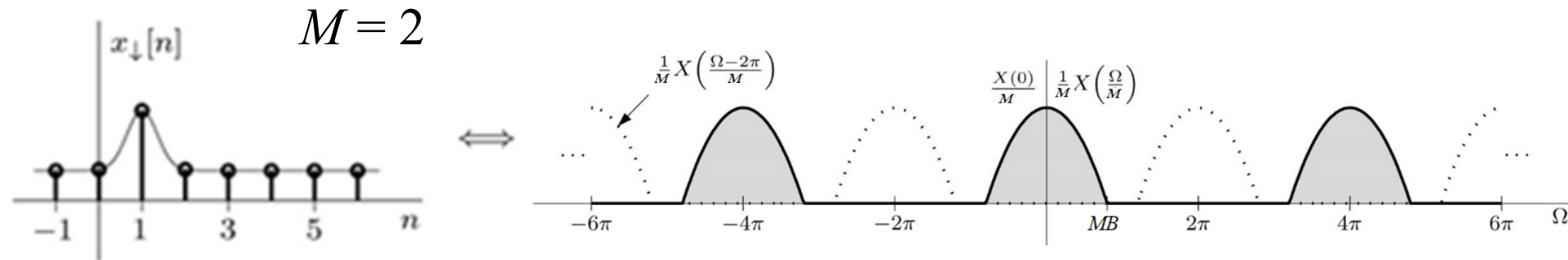
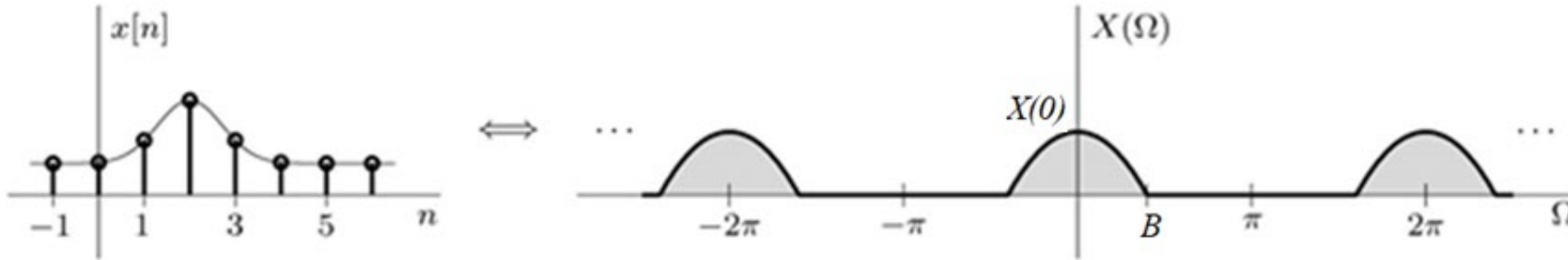


Interpolation

Down-Sampling



Decimation



$$x_\downarrow[n] = x[Mn]$$

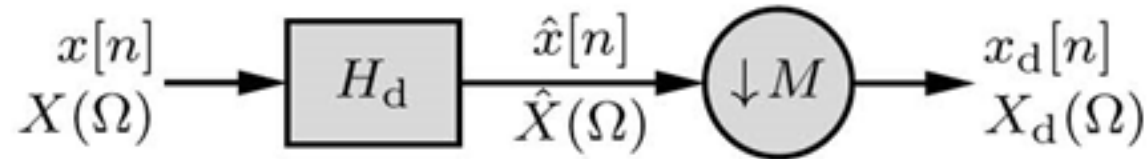
$$X_\downarrow(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} X\left(\frac{\Omega - 2\pi m}{M}\right)$$

To avoid aliasing $MB \leq \pi$ so take $x[n]$ to a low pass filter first with cutoff frequency π/M .

$$X_d(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} X\left(\frac{\Omega - 2\pi m}{M}\right) H_d\left(\frac{\Omega - 2\pi m}{M}\right)$$

Downsampling and Decimation

If the spectrum $X_{\downarrow}(\Omega)$ of the downsampled signal $x[n]$ does not meet this condition $MB \leq \pi$ then use a filter H_d with cutoff frequency $\Omega_c \leq \pi/M$.



$$\hat{X}(\Omega) = X(\Omega)H_d(\Omega)$$

Decimation

$$X_d(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} \hat{X}\left(\frac{\Omega - 2\pi m}{M}\right) = \frac{1}{M} \sum_{m=0}^{M-1} X\left(\frac{\Omega - 2\pi m}{M}\right) H_d\left(\frac{\Omega - 2\pi m}{M}\right) \quad \text{for } MB \leq \pi$$

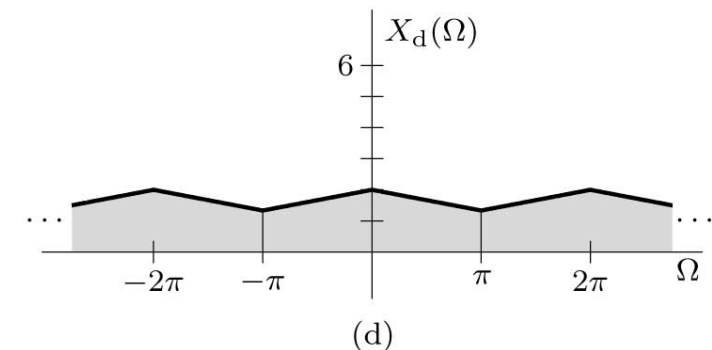
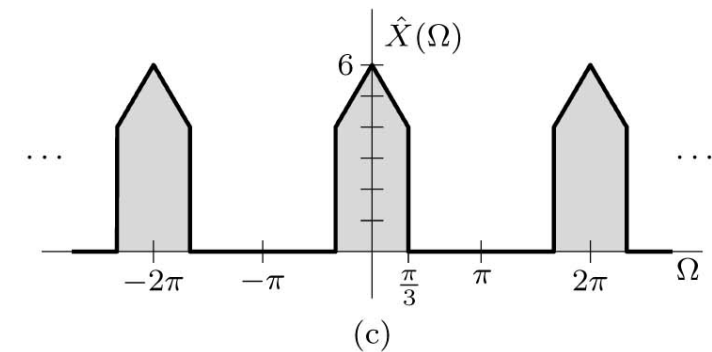
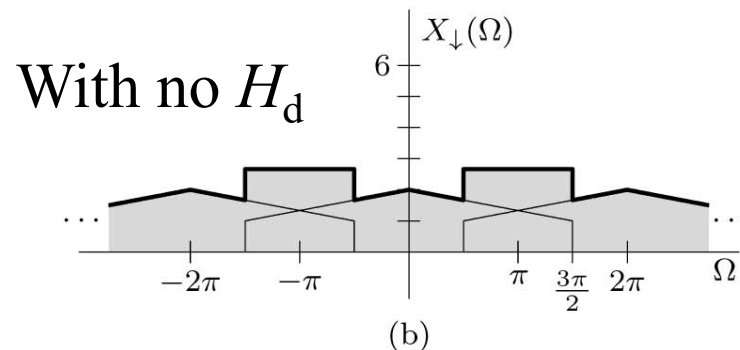
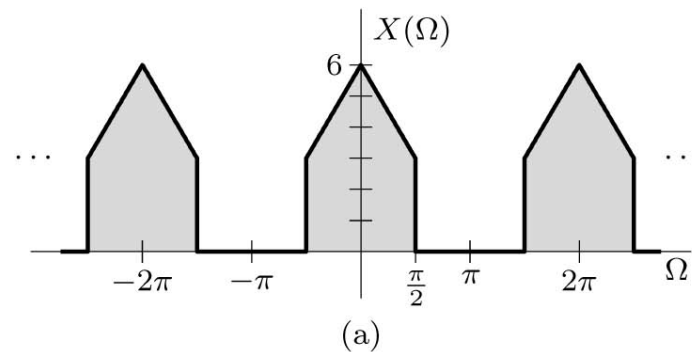
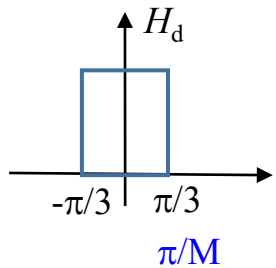
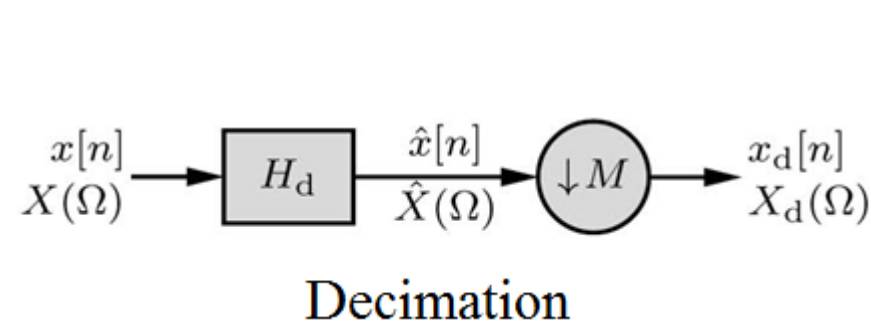
Time representation for Decimation

$$h_d[n] = \frac{1}{M} \text{sinc}(n/M) \qquad x_d[n] = \frac{1}{M} \sum_{k=-\infty}^{\infty} x[k] \text{sinc}\left(n - \frac{k}{M}\right)$$

Downsampling and Decimation

The DTFT $X(\Omega)$ of a signal $x[n]$ is shown below. Using $M = 3$, determine the DTFTs of the downsampled signal $x_{\downarrow}[n]$ and the decimated signal $x_d[n]$. Assume that the decimator uses an ideal lowpass filter.

$$X_{\downarrow}(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} X\left(\frac{\Omega - 2\pi m}{M}\right)$$



Downsampling and Decimation

A signal $x[n] = \gamma^n u[n]$, where $|\gamma| < 1$, is downsampled by factor $M = 2$. Determine the resulting downsampled signal $x_{\downarrow}[n] = x[2n]$ and its spectrum $X_{\downarrow}(\Omega)$.

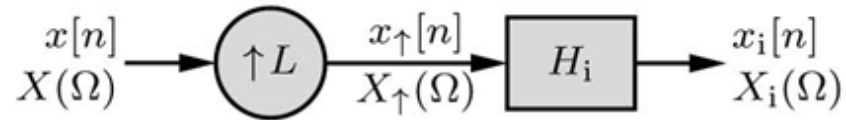
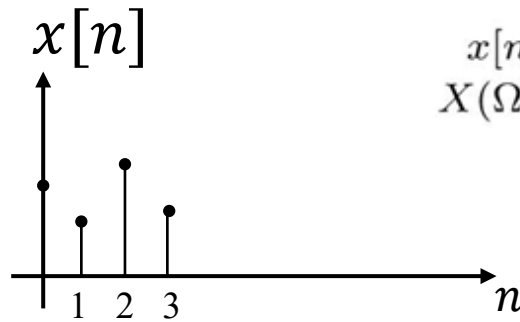
$$x[n] = \gamma^n u[n] \iff \frac{e^{j\Omega}}{e^{j\Omega} - \gamma} = \frac{1}{1 - \gamma e^{-j\Omega}} = X(\Omega)$$

$$x_{\downarrow}[n] = x[Mn] = x[2n] = \gamma^{2n} u[2n] = \gamma^{2n} u[n]$$

We can use the table to find the $X_{\downarrow}(\Omega)$ but let us use $X_{\downarrow}(\Omega) = \frac{1}{M} \sum_{m=0}^{M-1} X\left(\frac{\Omega - 2\pi m}{M}\right)$

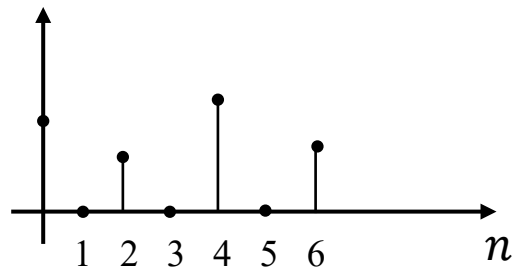
$$\begin{aligned} X_{\downarrow}(\Omega) &= \frac{1}{2} \left[X\left(\frac{\Omega}{2}\right) + X\left(\frac{\Omega - 2\pi}{2}\right) \right] \\ &= \frac{1}{2} \left[X\left(\frac{\Omega}{2}\right) + X\left(\frac{\Omega}{2} - \pi\right) \right] \\ &= \frac{1}{2(1 - \gamma e^{-j\Omega/2})} + \frac{1}{2(1 + \gamma e^{-j\Omega/2})} \\ &= \frac{1}{1 - \gamma^2 e^{-j\Omega}}. \end{aligned}$$

Interpolation and Upsampling



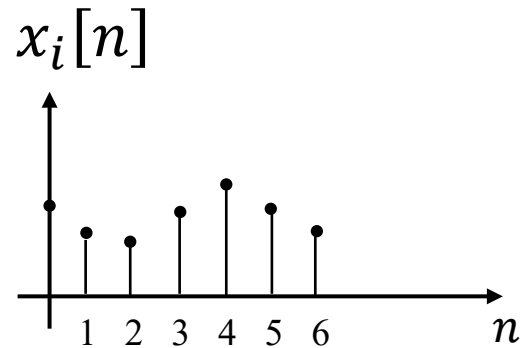
Interpolation

$$x_{\uparrow}[n] = x[n/L]$$

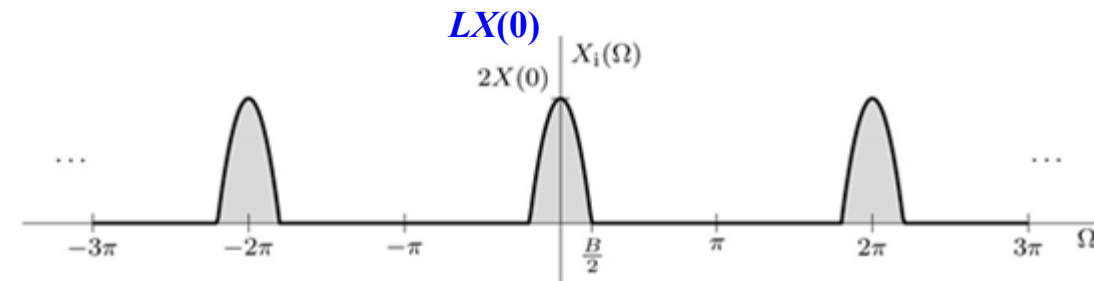
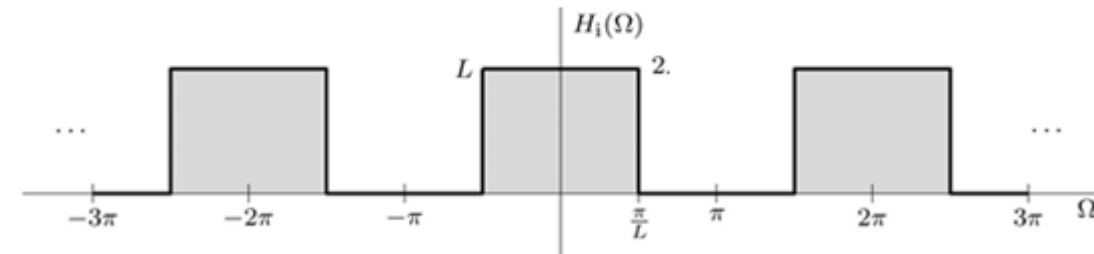
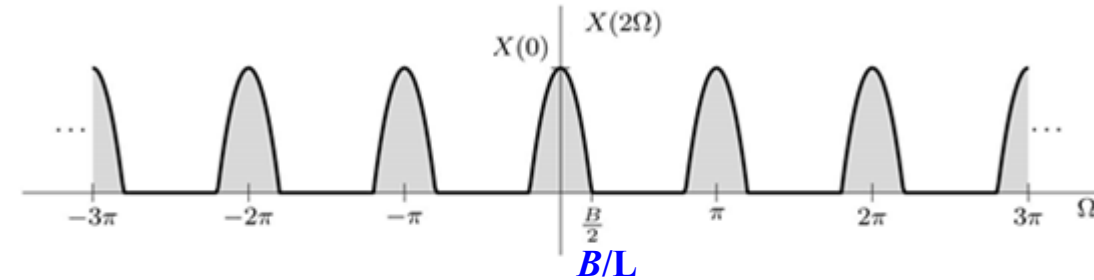
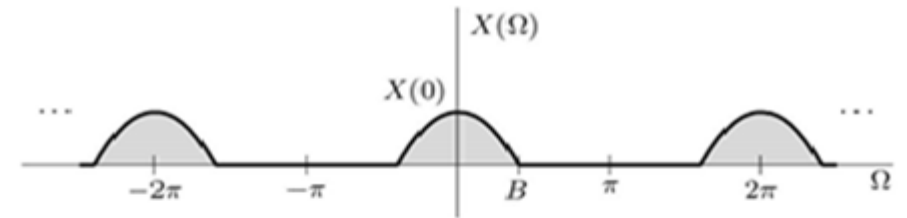


$$X_{\uparrow}(\Omega) = X(L\Omega)$$

$$H_i(\Omega) = L \sum_{k=-\infty}^{\infty} \Pi\left(\frac{\Omega - 2\pi k}{2\pi/L}\right)$$

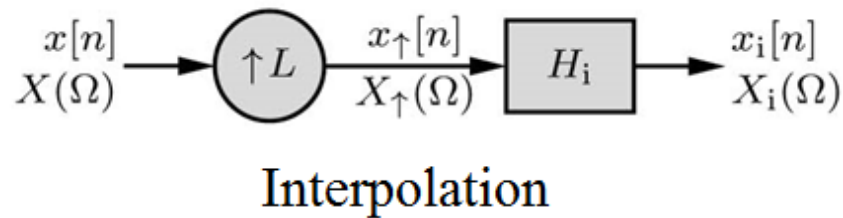


$$X_i(\Omega) = X_{\uparrow}(\Omega)H_i(\Omega)$$



Interpolation and Upsampling

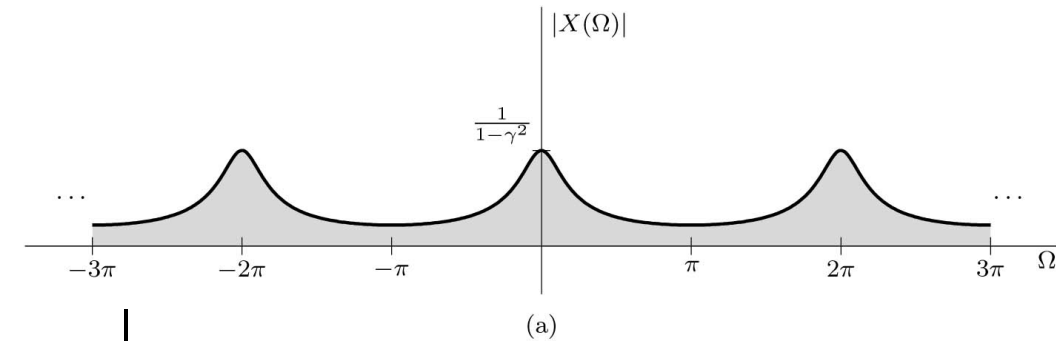
Consider a signal $x[n] = \gamma^{2n}u[n]$, where $|\gamma| < 1$.
Using $L = 2$, determine the spectra of the upsampled signal $x_{\uparrow}[n]$ and the ideally interpolated signal $x_i[n]$.



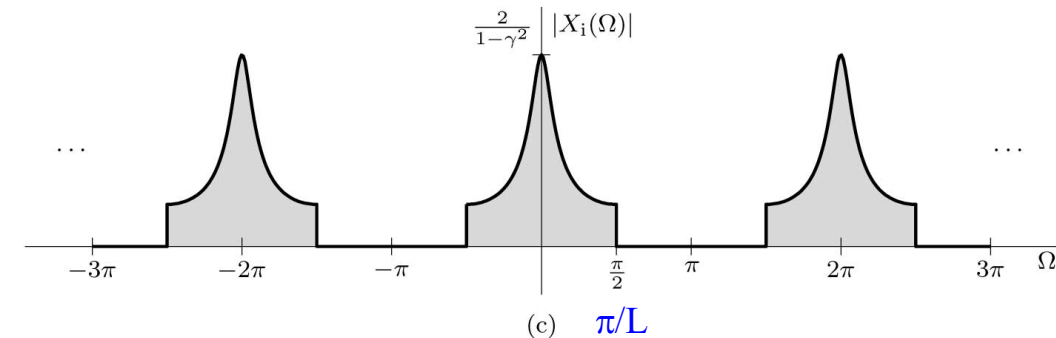
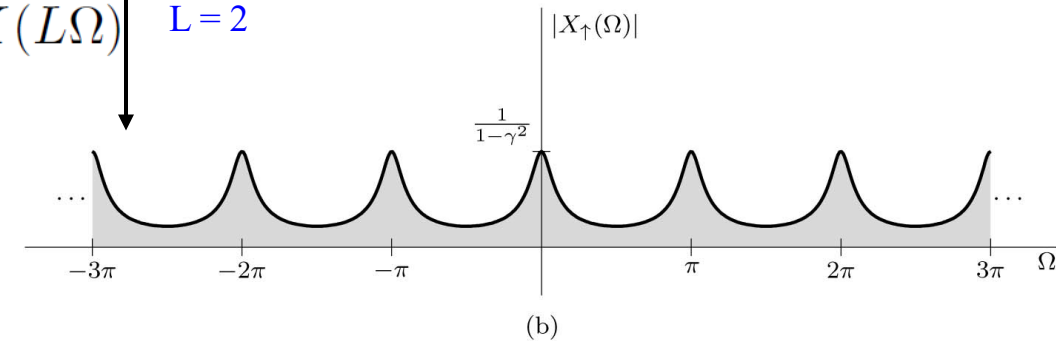
$$X(\Omega) = 1/(1 - \gamma^2 e^{-j\Omega})$$

$$X_{\uparrow}(\Omega) = X(2\Omega) = \frac{1}{1 - \gamma^2 e^{-j2\Omega}}$$

$$X_i(\Omega) = X_{\uparrow}(\Omega)H_i(\Omega) = \frac{2}{1 - \gamma^2 e^{-j2\Omega}} \Pi\left(\frac{\Omega}{\pi}\right), \quad |\Omega| \leq \pi$$

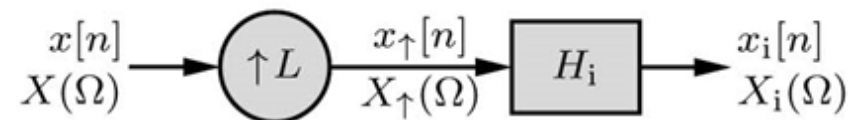


$$X_{\uparrow}(\Omega) = X(L\Omega) \quad L=2$$

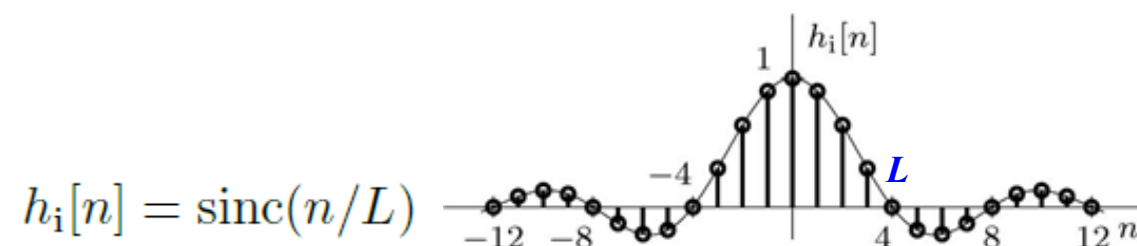
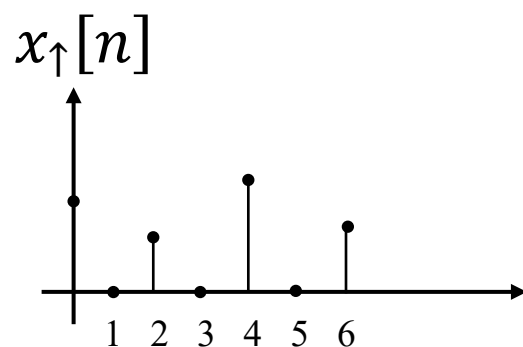


Time-Domain Characterization of Up-Sampling

$$x_i[n] = x_{\uparrow}[n] * h_i[n] = \sum_{m=-\infty}^{\infty} x_{\uparrow}[m] h_i[n - m]$$

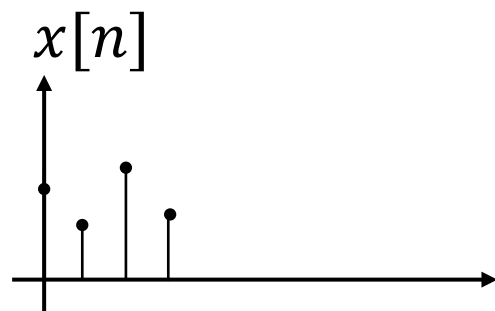
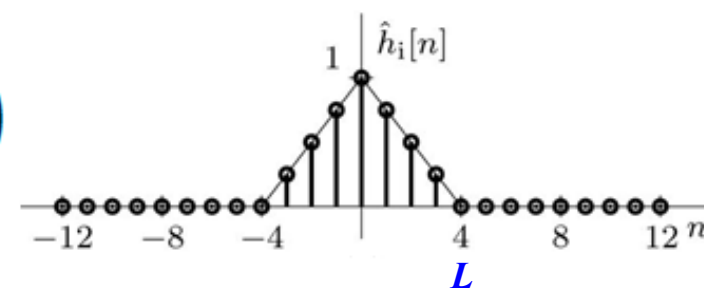


Interpolation



$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k] h_i[n - kL]$$

$$\hat{h}_i[n] = \Lambda\left(\frac{n}{2L}\right)$$



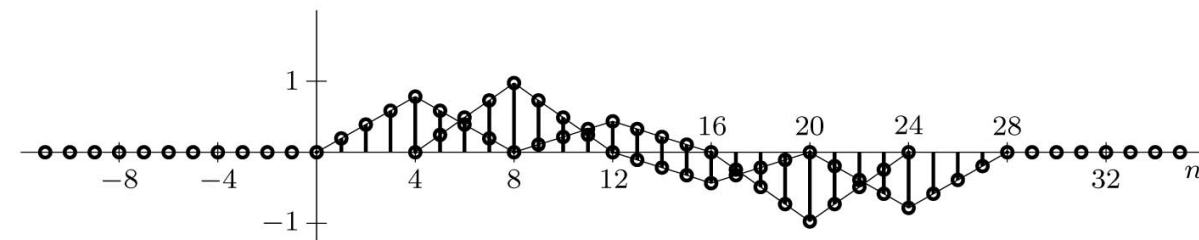
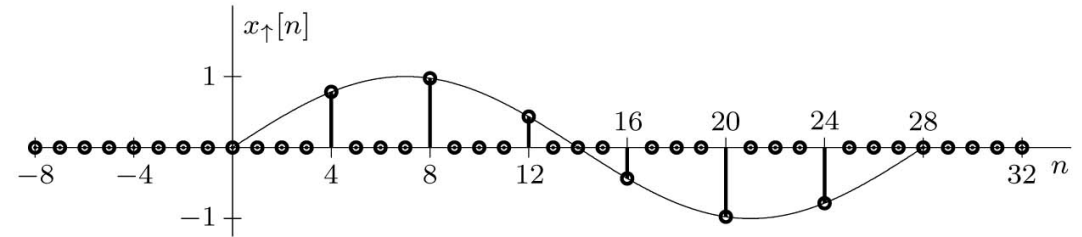
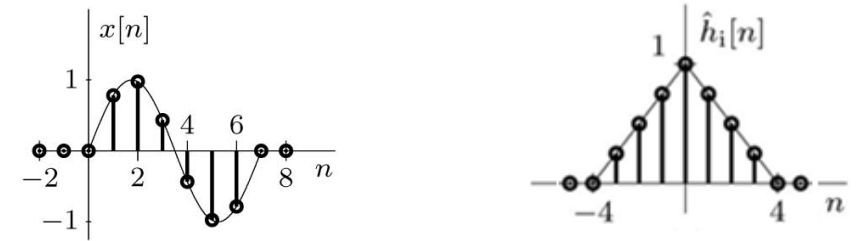
$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k] \text{sinc}\left(\frac{n - kL}{L}\right)$$

Example

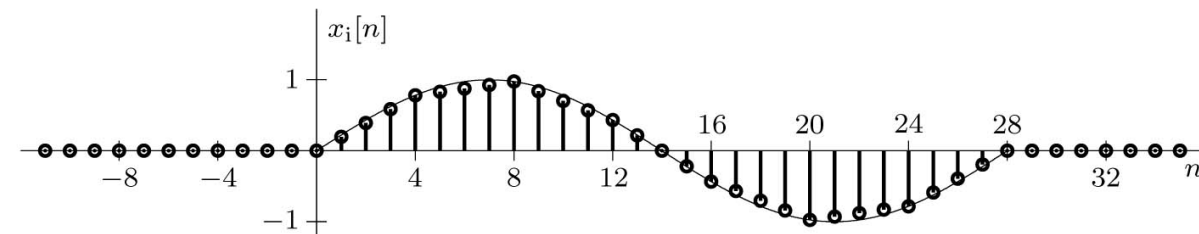
Using the $L = 4$ linear interpolator in Fig. below^(a) and an input of $x[n] = \sin(2\pi n/7)(u[n] - u[n-7])$, determine and sketch the interpolator output $x_i[n]$.

$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k]h_i[n - kL] \quad \hat{h}_i[n] = \Lambda\left(\frac{n}{8}\right) \quad (b)$$

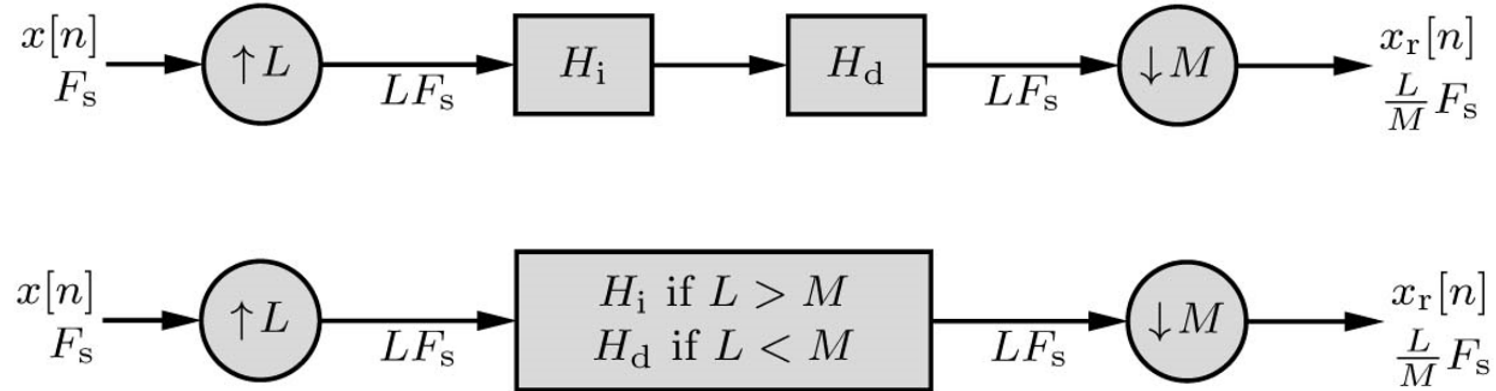
```
x = @(n)
sin(2*pi*n/7) .* ((n>=0) & (n<7)) .* (mod(n,1)==0);
n = -2:8;
subplot(311); stem(n,x(n)); xlabel('n');
ylabel('x[n]');
xup = @(n) x(n/4); n2 = (n(1)*4:n(end)*4);
subplot(312); stem(n2,xup(n2)); xlabel('n');
ylabel('x_{\uparrow}[n]');
hi = [1 2 3 4 3 2 1]/4;
n3 = (-3+n2(1):3+n2(end));
subplot(313); stem(n3,conv(xup(n2),hi));
xlabel('n'); ylabel('x_i[n]');
```



(d)



Fractional Sampling Rate Conversion



- An interpolator/decimator cascade used for fractional sampling rate changes.
- Upsampling by L followed by downsampling by M changes the overall sampling rate by a fractional amount L/M .
- The two lowpass filters H_i and H_d , being in cascade, can be replaced by a single lowpass filter of cutoff frequency π/L or π/M , whichever is lower.
- Preferable to do upsampling prior to downsampling to avoid loss of information.

Example: Suppose that the bandwidth of $x[n]$ is $\pi/3$, and we wish to change the sampling rate by factor $3/5$.

6.7

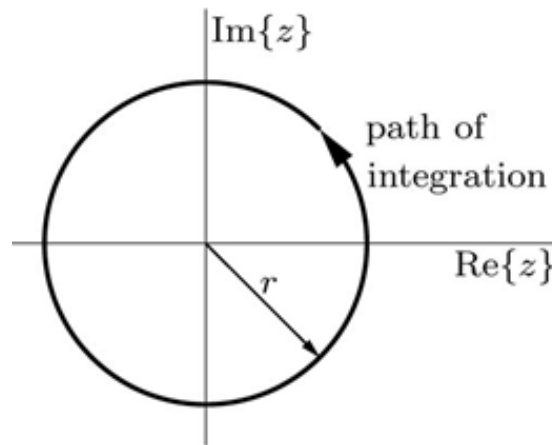
Generalization of the DTFT to the z-Transform

Generalization of the DTFT to the z-Transform

- 1) DTFT synthesizes an arbitrary signal $x[n]$ using sinusoids or complex exponentials of the form $e^{j\Omega n}$. DTFT exists only for absolutely summable signals.
- 2) In System analysis, DTFT is incapable of handling exponentially growing (or decaying) signals.
- 3) z-transform avoids the limitation of the DTFT by generalizing the complex frequency by replacing $j\Omega$ by $\sigma + j\Omega$, where Ω is the oscillation rate and σ is the decay (or grow) rate.

$$\begin{aligned} X(j\Omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \\ &\quad \swarrow j\Omega \rightarrow \sigma + j\Omega \\ &\quad \sum_{n=-\infty}^{\infty} x[n] e^{-(\sigma + j\Omega)n} \\ &\quad \swarrow z = e^{\sigma + j\Omega} \\ X(z) &= \sum_{n=-\infty}^{\infty} x[n] z^{-n} \end{aligned}$$

z-transform



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(j\Omega) e^{j\Omega n} d\Omega$$

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

Inverse z-transform